



THEORY OF MATRICES - (MMT126J)

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Chapter 1

MATRIX ALGEBRA AND STRUCTURAL PROPERTIES

This unit introduces the fundamental algebraic structure of matrices over the real field $\mathbb{R}$ and complex field $\mathbb{C}$. It examines core operational properties, highlighting key structural differences from scalar algebra, particularly non-commutativity. The unit further covers transpose and conjugate transpose operations, reversal laws, structural classifications such as symmetric, skew-symmetric, Hermitian, and skew-Hermitian matrices, and matrix decompositions. Finally, it explores the adjoint of a square matrix and its essential relationship with matrix inversion.

1.1 Algebraic Operations on Matrices over Real and Complex Fields

Definition 1.1.1 (Matrix over $\mathbb{R}$ and $\mathbb{C}$). Let $\mathbb{K}$ denote either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$. A matrix A of size $m \times n$ over $\mathbb{K}$ is a rectangular array of scalars arranged in m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = (a_{ij})_{m \times n}, \quad a_{ij} \in \mathbb{K}.$$

The set of all $m \times n$ matrices with entries in $\mathbb{K}$ is denoted by $M_{m \times n}(\mathbb{K})$ or $M_n(\mathbb{K})$ when $m = n$.

1.1.1 Basic Matrix Operations

Definition 1.1.2 (Matrix Addition and Scalar Multiplication). Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices in $M_{m \times n}(\mathbb{K})$, and let $\alpha \in \mathbb{K}$.

(i) **Matrix Addition:** The sum $A + B$ is an $m \times n$ matrix defined component-wise by:

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

Addition is defined *only* when A and B have identical dimensions.

(ii) **Scalar Multiplication:** The scalar multiple αA is an $m \times n$ matrix defined by:

$$(\alpha A)_{ij} = \alpha \cdot a_{ij} \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

Theorem 1.1.3 (Properties of Addition and Scalar Multiplication). *For any matrices $A, B, C \in M_{m \times n}(\mathbb{K})$ and scalars $\alpha, \beta \in \mathbb{K}$, the following properties hold:*

(i) *Commutativity of Addition:* $A + B = B + A$

- (ii) *Associativity of Addition:* $(A + B) + C = A + (B + C)$
- (iii) *Additive Identity:* $A + 0 = A$, where 0 is the $m \times n$ zero matrix.
- (iv) *Additive Inverse:* $A + (-A) = 0$, where $-A = (-1)A$.
- (v) *Distributive Laws:* $\alpha(A + B) = \alpha A + \alpha B$ and $(\alpha + \beta)A = \alpha A + \beta A$.
- (vi) *Compatibility of Scalar Multiplication:* $(\alpha\beta)A = \alpha(\beta A)$.

1.1.2 Matrix Multiplication

Definition 1.1.4 (Matrix Multiplication). Let $A = (a_{ij}) \in M_{m \times p}(\mathbb{K})$ and $B = (b_{jk}) \in M_{p \times n}(\mathbb{K})$. The product AB is the $m \times n$ matrix $C = (c_{ik}) \in M_{m \times n}(\mathbb{K})$ whose (i, k) -th entry is given by:

$$c_{ik} = (AB)_{ik} = \sum_{j=1}^p a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{ip}b_{pk}.$$

The product AB is defined if and only if the number of columns in A equals the number of rows in B .

1.1.3 Non-Commutativity of Matrix Multiplication

Unlike scalar multiplication in $\mathbb{R}$ or $\mathbb{C}$ where $ab = ba$, matrix multiplication is **generally non-commutative**. That is, for two matrices A and B : $AB \neq BA$ in general.

Non-commutativity manifests in three distinct scenarios:

1. **Dimensional Mismatch:** AB is defined, but BA is not defined (e.g., $A \in M_{2 \times 3}(\mathbb{K})$ and $B \in M_{3 \times 4}(\mathbb{K})$).
2. **Dimension Discrepancy:** Both AB and BA exist, but they have different dimensions (e.g., $A \in M_{2 \times 3}(\mathbb{K})$ and $B \in M_{3 \times 2}(\mathbb{K})$ yield $AB \in M_{2 \times 2}(\mathbb{K})$ and $BA \in M_{3 \times 3}(\mathbb{K})$).
3. **Entry-wise Inequality:** $A, B \in M_n(\mathbb{K})$ (both are square matrices of the same order), so both AB and BA are $n \times n$ matrices, yet $AB \neq BA$.

Example 1.1.5. Consider the real matrices $A, B \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Computing AB :

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1(0) + 2(1) & 1(1) + 2(0) \\ 3(0) + 4(1) & 3(1) + 4(0) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}.$$

Computing BA :

$$BA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0(1) + 1(3) & 0(2) + 1(4) \\ 1(1) + 0(3) & 1(2) + 0(4) \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}.$$

Clearly, $AB \neq BA$.

Example 1.1.6. Consider the complex matrices $A, B \in M_2(\mathbb{C})$:

$$A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Calculating AB :

$$AB = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}.$$

Calculating BA :

$$BA = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}.$$

Notice that $AB = -BA \neq BA$. Thus, A and B anti-commute.

1.1.4 Consequences of Non-Commutativity

The non-commutative nature of matrix multiplication leads to crucial algebraic differences between matrix algebra and scalar arithmetic:

Failure of the binomial formula. In standard real or complex algebra, $(a + b)^2 = a^2 + 2ab + b^2$. However, for square matrices A and B :

$$(A + B)^2 = (A + B)(A + B) = A^2 + AB + BA + B^2.$$

Because $AB \neq BA$ in general:

$$(A + B)^2 \neq A^2 + 2AB + B^2 \quad (\text{unless } AB = BA).$$

Similarly:

$$(A - B)(A + B) = A^2 + AB - BA - B^2 \neq A^2 - B^2 \quad (\text{unless } AB = BA).$$

Example 1.1.7. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $A + B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

$$(A + B)^2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}.$$

On the other hand:

$$A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad AB = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

So $A^2 + 2AB + B^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \neq (A + B)^2$.

Zero products.

Example 1.1.8. Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \neq 0$.

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0.$$

Failure of cancellation. If $AB = AC$, it does **not** generally follow that $B = C$, even if $A \neq 0$.

Example 1.1.9. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$, and $C = \begin{bmatrix} 0 & 0 \\ 3 & 4 \end{bmatrix}$. Notice that $B \neq C$. However:

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$AC = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus $AB = AC = 0$, yet $B \neq C$.

Remark 1.1.10. The cancellation law $AB = AC \implies B = C$ holds if and only if A is non-singular (invertible).

1.1.5 Special Cases Where Matrices Commute

Although matrix multiplication is non-commutative in general, specific pairs of matrices commute ($AB = BA$):

- (i) **Identity Matrix:** Any square matrix $A \in M_n(\mathbb{K})$ commutes with the identity matrix I_n : $AI_n = I_nA = A$.
- (ii) **Scalar Matrices:** Matrices of the form cI_n (where $c \in \mathbb{K}$) commute with all matrices in $M_n(\mathbb{K})$.
- (iii) **Powers of the Same Matrix:** For any $k, m \in \mathbb{N}_0$, $A^k A^m = A^m A^k = A^{k+m}$.
- (iv) **Matrix and Its Inverse:** If A is invertible, $AA^{-1} = A^{-1}A = I_n$.
- (v) **Diagonal Matrices:** Two diagonal matrices of the same size always commute with each other.

1.2 Transpose of a Matrix and Its Properties

Definition 1.2.1 (Transpose of a Matrix). Let $A = (a_{ij})$ be an $m \times n$ matrix over $\mathbb{K}$. The **transpose** of A , denoted by A^T (or A^t), is the $n \times m$ matrix obtained by interchanging the rows and columns of A .

Formally, if $A^T = (b_{ji})_{n \times m}$, then its entries are defined by:

$$b_{ji} = a_{ij} \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

Thus, the i -th row of A becomes the i -th column of A^T , and the j -th column of A becomes the j -th row of A^T .

1.2.1 Algebraic Properties of Transpose

Theorem 1.2.2 (Properties of Transposition). Let A and B be matrices of appropriate sizes over $\mathbb{K}$, and let $\alpha \in \mathbb{K}$. Then:

- (i) **Double Transpose Law:** $(A^T)^T = A$.
- (ii) **Sum Rule:** $(A + B)^T = A^T + B^T$, where A and B have the same dimensions.
- (iii) **Scalar Multiplication Rule:** $(\alpha A)^T = \alpha A^T$.
- (iv) **Reversal Law for Multiplication:** $(AB)^T = B^T A^T$, provided the product AB is defined.

Proof. (i) Let $A = (a_{ij})_{m \times n}$. Then $A^T = (b_{ji})_{n \times m}$ where $b_{ji} = a_{ij}$.

Taking the transpose again, $(A^T)^T = (c_{ij})_{m \times n}$ where $c_{ij} = b_{ji} = a_{ij}$.

Thus, $(A^T)^T = A$.

- (ii) Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size. The (i, j) -th entry of $(A + B)^T$ is $(A + B)_{ji} = a_{ji} + b_{ji}$. This is also the (i, j) -th entry of $A^T + B^T$. Hence, $(A + B)^T = A^T + B^T$.

- (iii) Let $A = (a_{ij})$. The (i, j) -th entry of $(\alpha A)^T$ is $(\alpha A)_{ji} = \alpha a_{ji}$. This is also the (i, j) -th entry of αA^T . Hence, $(\alpha A)^T = \alpha A^T$.
- (iv) Let $A = (a_{ij})_{m \times p}$ and $B = (b_{jk})_{p \times n}$. Then AB is an $m \times n$ matrix. The (i, k) -th entry of AB is given by:

$$(AB)_{ik} = \sum_{j=1}^p a_{ij} b_{jk}.$$

Therefore, the (k, i) -th entry of $(AB)^T$ is:

$$\left((AB)^T\right)_{ki} = (AB)_{ik} = \sum_{j=1}^p a_{ij} b_{jk}.$$

Now, consider $B^T = (d_{kj})_{n \times p}$ where $d_{kj} = b_{jk}$, and $A^T = (e_{ji})_{p \times m}$ where $e_{ji} = a_{ij}$.

The (k, i) -th entry of the product $B^T A^T$ is:

$$(B^T A^T)_{ki} = \sum_{j=1}^p d_{kj} e_{ji} = \sum_{j=1}^p b_{jk} a_{ij} = \sum_{j=1}^p a_{ij} b_{jk}.$$

Since $\left((AB)^T\right)_{ki} = (B^T A^T)_{ki}$ for all $1 \leq k \leq n$ and $1 \leq i \leq m$, it follows that $(AB)^T = B^T A^T$. ■

Theorem 1.2.3 (General Reversal Law for Transposition). *Let $k \geq 2$ be an integer, and let $A_1, A_2, \dots, A_k$ be a finite collection of conformable matrices over $\mathbb{K}$. Then the transpose of their product satisfies:*

$$(A_1 A_2 \cdots A_k)^T = A_k^T \cdots A_2^T A_1^T.$$

Proof. We prove the statement by mathematical induction on k , the number of matrices.

Base Case ($k = 2$): For two conformable matrices A_1 and A_2 , the reversal law gives: $(A_1 A_2)^T = A_2^T A_1^T$. Thus, the statement holds for $k = 2$.

Inductive Hypothesis: Assume that the statement holds for $k = m$ matrices for some integer $m \geq 2$. That is, assume:

$$(A_1 A_2 \cdots A_m)^T = A_m^T \cdots A_2^T A_1^T.$$

Inductive Step ($k = m+1$): Let $A_1, A_2, \dots, A_m, A_{m+1}$ be $m+1$ conformable matrices. Group the product of the first m matrices into a single matrix $B = A_1 A_2 \cdots A_m$.

Then, the product of all $m+1$ matrices can be written as BA_{m+1} . Applying the base case ($k = 2$) to the product of B and A_{m+1} , we obtain:

$$(A_1 A_2 \cdots A_m A_{m+1})^T = (BA_{m+1})^T = A_{m+1}^T B^T.$$

By the inductive hypothesis, $B^T = (A_1 A_2 \cdots A_m)^T = A_m^T \cdots A_2^T A_1^T$. Substituting this expression for B^T gives:

$$(A_1 A_2 \cdots A_m A_{m+1})^T = A_{m+1}^T (A_m^T \cdots A_2^T A_1^T) = A_{m+1}^T A_m^T \cdots A_2^T A_1^T.$$

Hence, the statement holds for $k = m+1$.

By the principle of mathematical induction, the general reversal law $(A_1 A_2 \cdots A_k)^T = A_k^T \cdots A_2^T A_1^T$ holds for all integers $k \geq 2$. ■

Example 1.2.4. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix}$ and $B^T = \begin{bmatrix} 2 & 0 & 1 \\ 1 & -1 & 3 \end{bmatrix}$.

Computing AB :

$$AB = \begin{bmatrix} 1(2) + 2(0) + 0(1) & 1(1) + 2(-1) + 0(3) \\ 3(2) + (-1)(0) + 4(1) & 3(1) + (-1)(-1) + 4(3) \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 16 \end{bmatrix}.$$

Thus, $(AB)^T = \begin{bmatrix} 2 & 10 \\ -1 & 16 \end{bmatrix}$.

Now computing $B^T A^T$:

$$B^T A^T = \begin{bmatrix} 2 & 0 & 1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2(1) + 0(2) + 1(0) & 2(3) + 0(-1) + 1(4) \\ 1(1) + (-1)(2) + 3(0) & 1(3) + (-1)(-1) + 3(4) \end{bmatrix} = \begin{bmatrix} 2 & 10 \\ -1 & 16 \end{bmatrix}.$$

This confirms $(AB)^T = B^T A^T$.

1.3 Inverse of a Matrix and Its Properties

Definition 1.3.1 (Invertible Matrix and Inverse). A square matrix $A \in M_n(\mathbb{K})$ is said to be **invertible** (or **non-singular**) if there exists a matrix $B \in M_n(\mathbb{K})$ such that: $AB = BA = I_n$, where I_n is the $n \times n$ identity matrix.

If such a matrix B exists, it is unique and is called the **inverse** of A , denoted by A^{-1} .

Theorem 1.3.2 (Uniqueness of Inverse). If a square matrix $A \in M_n(\mathbb{K})$ is invertible, its inverse is unique.

Proof. Suppose B and C are both inverses of A . Then:

$$AB = BA = I_n \quad \text{and} \quad AC = CA = I_n.$$

Using associativity of matrix multiplication: $B = BI_n = B(AC) = (BA)C = I_n C = C$. Hence, $B = C$. ■

1.3.1 Properties of Inverse and Reversal Laws

Theorem 1.3.3 (Properties of Matrix Inverses). Let $A, B \in M_n(\mathbb{K})$ be invertible matrices, and let $\alpha \in \mathbb{K} \setminus \{0\}$. Then:

(i) **Involution Law:** A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(ii) **Scalar Multiplication Law:** αA is invertible and $(\alpha A)^{-1} = \frac{1}{\alpha} A^{-1}$.

(iii) **Reversal Law for Inverse:** The product AB is invertible and $(AB)^{-1} = B^{-1} A^{-1}$.

(iv) **Commutativity of Transpose and Inverse:** A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

Proof. (i) Since $AA^{-1} = A^{-1}A = I_n$, the matrix A is the inverse of A^{-1} . Hence $(A^{-1})^{-1} = A$.

(ii) We have

$$(\alpha A) \left(\frac{1}{\alpha} A^{-1} \right) = \left(\frac{1}{\alpha} A^{-1} \right) (\alpha A) = I_n.$$

Therefore, $(\alpha A)^{-1} = \frac{1}{\alpha} A^{-1}$.

(iii) Observe that $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = I_n$, and $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = I_n$. Hence $(AB)^{-1} = B^{-1}A^{-1}$.

(iv) Since $AA^{-1} = I_n$, taking transpose gives $(A^{-1})^T A^T = I_n$. Similarly, from $A^{-1}A = I_n$, $A^T(A^{-1})^T = I_n$. Therefore $(A^{-1})^T$ is the inverse of A^T , so $(A^T)^{-1} = (A^{-1})^T$. ■

Theorem 1.3.4 (General Reversal Law for Inverses). *Let $k \geq 2$ be an integer, and let $A_1, A_2, \dots, A_k \in M_n(\mathbb{K})$ be invertible matrices. Then the product matrix $A_1 A_2 \cdots A_k$ is invertible, and its inverse satisfies:*

$$(A_1 A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1} A_1^{-1}.$$

Proof. We prove the theorem by mathematical induction on k , the number of invertible matrices.

Base Case ($k = 2$): For two invertible matrices $A_1, A_2 \in M_n(\mathbb{K})$, we verify that $(A_1 A_2)(A_2^{-1} A_1^{-1}) = I_n$ and $(A_2^{-1} A_1^{-1})(A_1 A_2) = I_n$:

$$(A_1 A_2)(A_2^{-1} A_1^{-1}) = A_1(A_2 A_2^{-1})A_1^{-1} = A_1 I_n A_1^{-1} = A_1 A_1^{-1} = I_n,$$

and

$$(A_2^{-1} A_1^{-1})(A_1 A_2) = A_2^{-1}(A_1^{-1} A_1)A_2 = A_2^{-1} I_n A_2 = A_2^{-1} A_2 = I_n.$$

By uniqueness of the matrix inverse, $(A_1 A_2)^{-1} = A_2^{-1} A_1^{-1}$. Thus, the base case holds.

Inductive Hypothesis: Assume that the statement holds for $k = m$ invertible matrices for some integer $m \geq 2$. That is, assume $A_1 A_2 \cdots A_m$ is invertible and:

$$(A_1 A_2 \cdots A_m)^{-1} = A_m^{-1} \cdots A_2^{-1} A_1^{-1}.$$

Inductive Step ($k = m + 1$): Let $A_1, A_2, \dots, A_m, A_{m+1} \in M_n(\mathbb{K})$ be $m + 1$ invertible matrices. Group the product of the first m matrices as $B = A_1 A_2 \cdots A_m$.

Since B is invertible by the inductive hypothesis and A_{m+1} is invertible by assumption, we apply the base case ($k = 2$) to the product BA_{m+1} :

$$(A_1 A_2 \cdots A_m A_{m+1})^{-1} = (BA_{m+1})^{-1} = A_{m+1}^{-1} B^{-1}.$$

Substituting the inductive hypothesis $B^{-1} = (A_1 A_2 \cdots A_m)^{-1} = A_m^{-1} \cdots A_2^{-1} A_1^{-1}$ yields:

$$(A_1 A_2 \cdots A_m A_{m+1})^{-1} = A_{m+1}^{-1} (A_m^{-1} \cdots A_2^{-1} A_1^{-1}) = A_{m+1}^{-1} A_m^{-1} \cdots A_2^{-1} A_1^{-1}.$$

Thus, the statement holds for $k = m + 1$.

By the principle of mathematical induction, the general reversal law for inverses holds for all integers $k \geq 2$. ■

1.4 Adjoint of a Square Matrix and Matrix Inversion

The classical adjoint (also known as the **adjugate**) of a square matrix provides an explicit formula for matrix inversion and bridges matrix algebra with determinant theory.

1.4.1 Minors, Cofactors, and the Adjoint Matrix

Definition 1.4.1 (Minor and Cofactor). Let $A = (a_{ij}) \in M_n(\mathbb{K})$ be a square matrix of order $n \geq 2$.

- (i) The **minor** M_{ij} of the entry a_{ij} is the determinant of the $(n - 1) \times (n - 1)$ submatrix obtained by deleting the i -th row and j -th column of A .

- (ii) The **cofactor** C_{ij} of the entry a_{ij} is defined as: $C_{ij} = (-1)^{i+j}M_{ij}$. The matrix $C = (C_{ij})_{n \times n}$ is called the **cofactor matrix** of A .

Definition 1.4.2 (Adjoint of a Matrix). Let $A \in M_n(\mathbb{K})$ be a square matrix and let $C = (C_{ij})$ be its cofactor matrix. The **adjoint** (or **adjugate**) of A , denoted by $\text{adj}(A)$, is the transpose of the cofactor matrix:

$$\text{adj}(A) = C^T = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

Explicitly, the (i, j) -th entry of $\text{adj}(A)$ is the cofactor C_{ji} .

1.4.2 Fundamental Theorem of Adjoint Matrix

The connection between A , $\text{adj}(A)$, and $\det(A)$ is governed by Laplace's expansion theorem.

Theorem 1.4.3 (Fundamental Property of Adjoint). For any square matrix $A \in M_n(\mathbb{K})$,

$$A \text{adj}(A) = \text{adj}(A)A = |A|I_n.$$

Proof. Let C_{ij} denote the cofactor of a_{ij} . Since

$$\text{adj}(A) = (C_{ji}),$$

the (i, k) -th entry of $A \text{adj}(A)$ is

$$\sum_{j=1}^n a_{ij} C_{kj}.$$

If $i = k$, then

$$\sum_{j=1}^n a_{ij} C_{ij} = |A|$$

by the Laplace expansion of $|A|$ along the i -th row.

If $i \neq k$, then

$$\sum_{j=1}^n a_{ij} C_{kj} = 0,$$

because it is the determinant obtained by replacing the k -th row of A by its i -th row; the resulting matrix has two equal rows.

Hence every diagonal entry of $A \text{adj}(A)$ is $|A|$, and every off-diagonal entry is 0. Therefore,

$$A \text{adj}(A) = |A|I_n.$$

Similarly, using Laplace expansion along columns,

$$\text{adj}(A)A = |A|I_n.$$

Thus,

$$A \text{adj}(A) = \text{adj}(A)A = |A|I_n.$$



1.4.3 Connection with Matrix Inversion

Theorem 1.4.4 (Criterion for Invertibility and Adjoint Formula). *A square matrix $A \in M_n(\mathbb{K})$ is invertible (non-singular) if and only if $|A| \neq 0$. In this case,*

$$A^{-1} = \frac{1}{|A|} \text{adj}(A).$$

Proof. Suppose first that $|A| \neq 0$. By the fundamental identity for the adjoint,

$$A \text{adj}(A) = \text{adj}(A)A = |A|I_n.$$

Since $|A| \neq 0$, we may divide by $|A|$ to obtain

$$A \left(\frac{1}{|A|} \text{adj}(A) \right) = \left(\frac{1}{|A|} \text{adj}(A) \right) A = I_n.$$

Thus A is invertible, and by the uniqueness of the inverse,

$$A^{-1} = \frac{1}{|A|} \text{adj}(A).$$

Conversely, suppose that A is invertible. Then $AA^{-1} = I_n$. Taking determinants gives $|A| |A^{-1}| = |I_n| = 1$. Hence $|A| \neq 0$. Therefore, A is invertible if and only if $|A| \neq 0$. ■

1.4.4 Further Properties of the Adjoint Matrix

Remark 1.4.5 (Additional Material). The core syllabus requires the basic idea of the adjoint and its connection with matrix inversion. The identities in this subsection are useful extensions and may be read after the fundamental formula $A^{-1} = (\det A)^{-1} \text{adj}(A)$ has been understood.

Theorem 1.4.6 (Algebraic Properties of Adjoint). *Let $A, B \in M_n(\mathbb{K})$ and $\alpha \in \mathbb{K}$. Then:*

1. $\text{adj}(I_n) = I_n$.
2. $\text{adj}(A^T) = (\text{adj}(A))^T$.
3. **Reversal Law:** $\text{adj}(AB) = \text{adj}(B) \text{adj}(A)$.
[Proof of singular case is not recommended at this stage.]
4. $\text{adj}(\alpha A) = \alpha^{n-1} \text{adj}(A)$.
5. **Determinant of Adjoint:** $\det(\text{adj}(A)) = (\det A)^{n-1}$.
6. **Double Adjoint Identity:** $\text{adj}(\text{adj}(A)) = (\det A)^{n-2} A, \quad n \geq 2$.

Proof. We use the fundamental identity

$$A \text{adj}(A) = \text{adj}(A)A = |A|I_n.$$

In particular, if A is invertible, then

$$\text{adj}(A) = |A|A^{-1}.$$

(i) Since $|I_n| = 1$ and $I_n^{-1} = I_n$, we have

$$\text{adj}(I_n) = |I_n|I_n^{-1} = I_n.$$

(ii) By definition, $\text{adj}(A)$ is the transpose of the cofactor matrix of A . Transposing A interchanges its rows and columns, and hence interchanges the corresponding cofactors. Therefore

$$\text{adj}(A^T) = (\text{adj}(A))^T.$$

(iii) First suppose that A and B are invertible. Then

$$\begin{aligned}\text{adj}(AB) &= |AB|(AB)^{-1} \\ &= |A||B|B^{-1}A^{-1} \\ &= (|B|B^{-1})(|A|A^{-1}) \\ &= \text{adj}(B)\text{adj}(A).\end{aligned}$$

Now suppose that A or B is singular. Consider $A_t = A + tI_n$, $B_t = B + tI_n$. For all but finitely many values of t , both A_t and B_t are invertible. Hence, for such t ,

$$\text{adj}(A_t B_t) = \text{adj}(B_t) \text{adj}(A_t).$$

Both sides are matrices whose entries are polynomials in t . Since these polynomials agree for infinitely many values of t , they are identical. Putting $t = 0$, we obtain

$$\text{adj}(AB) = \text{adj}(B) \text{adj}(A).$$

(iv) Each entry of $\text{adj}(A)$ is, up to sign, the determinant of an $(n-1) \times (n-1)$ minor of A . Multiplying A by α multiplies each such minor by α^{n-1} . Hence

$$\text{adj}(\alpha A) = \alpha^{n-1} \text{adj}(A).$$

(v) From

$$A \text{adj}(A) = |A|I_n,$$

taking determinants gives

$$|A| |\text{adj}(A)| = ||A|I_n| = |A|^n.$$

If $|A| \neq 0$, dividing by $|A|$ gives

$$|\text{adj}(A)| = |A|^{n-1}.$$

If $|A| = 0$, then $\text{rank}(A) \leq n-1$. If $\text{rank}(A) \leq n-2$, all $(n-1)$ -order minors are zero, so $\text{adj}(A) = 0$. If $\text{rank}(A) = n-1$, then $\text{rank}(\text{adj}(A)) = 1$. Thus, in either case,

$$|\text{adj}(A)| = 0 = |A|^{n-1}.$$

Therefore

$$|\text{adj}(A)| = |A|^{n-1}.$$

(vi) First suppose that A is invertible. By part (v),

$$\begin{aligned}\text{adj}(\text{adj}(A)) &= |\text{adj}(A)| (\text{adj}(A))^{-1} \\ &= |A|^{n-1} (|A|A^{-1})^{-1} \\ &= |A|^{n-1} \frac{1}{|A|} A \\ &= |A|^{n-2} A.\end{aligned}$$

Now suppose that A is singular. If $n > 2$, then $|A|^{n-2} = 0$. Also, $\text{rank}(\text{adj}(A)) \leq 1$, so all minors of order $n-1 \geq 2$ of $\text{adj}(A)$ are zero. Hence

$$\text{adj}(\text{adj}(A)) = 0 = |A|^{n-2} A.$$

For $n = 2$, direct calculation gives

$$\text{adj}(\text{adj}(A)) = A.$$

Since $|A|^0 = 1$, we again have

$$\text{adj}(\text{adj}(A)) = |A|^{n-2}A.$$

Thus, for $n \geq 2$,

$$\text{adj}(\text{adj}(A)) = |A|^{n-2}A.$$

■

Example 1.4.7. Find the adjoint and inverse of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ over $\mathbb{R}$.

Solution. 1. Compute cofactors:

$$C_{11} = (-1)^{1+1}(d) = d, \quad C_{12} = (-1)^{1+2}(c) = -c,$$

$$C_{21} = (-1)^{2+1}(b) = -b, \quad C_{22} = (-1)^{2+2}(a) = a.$$

2. The cofactor matrix is $C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$. Taking the transpose gives:

$$\text{adj}(A) = C^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Rule of thumb for 2×2 matrices: Swap diagonal elements and negate off-diagonal elements.

3. Determinant: $\det(A) = ad - bc$.

4. If $ad - bc \neq 0$, then:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

■

Example 1.4.8. Find $\text{adj}(A)$ and A^{-1} for the real matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}.$$

Solution. Compute $\det(A)$ using expansion along Row 1:

$$\det(A) = 1 \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} - 2 \begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} + 3 \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix}$$

$$\det(A) = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1.$$

Since $\det(A) = 1 \neq 0$, A is invertible.

The cofactors are:

$$C_{11} = + \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} = -24, \quad C_{12} = - \begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} = 20, \quad C_{13} = + \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = -5,$$

$$C_{21} = - \begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} = 18, \quad C_{22} = + \begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} = -15, \quad C_{23} = - \begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} = 4,$$

$$C_{31} = + \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 5, \quad C_{32} = - \begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} = -4, \quad C_{33} = + \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1.$$

The cofactor matrix is:

$$C = \begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{bmatrix},$$

and so the adjoint is:

$$\text{adj}(A) = C^T = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}.$$

Finally,

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{1} \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}.$$

■

Example 1.4.9. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$. Compute A^{-1} , B^{-1} , and $(AB)^{-1}$, and verify that $(AB)^{-1} = B^{-1}A^{-1}$.

Solution. First, compute A^{-1} and B^{-1} :

$$A^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 1 \end{bmatrix}.$$

Now compute AB :

$$AB = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix}.$$

The inverse of AB is:

$$(AB)^{-1} = \frac{1}{4(1) - 2(1)} \begin{bmatrix} 1 & -2 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 1/2 & -1 \\ -1/2 & 2 \end{bmatrix}.$$

Computing $B^{-1}A^{-1}$:

$$B^{-1}A^{-1} = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & -1 \\ -1/2 & 2 \end{bmatrix}.$$

This verifies that $(AB)^{-1} = B^{-1}A^{-1}$.

■

1.5 Conjugate Transpose and Its Algebraic Properties

When dealing with complex matrices $M_{m \times n}(\mathbb{C})$, taking the simple transpose is often insufficient because complex multiplication involves complex conjugation.

Definition 1.5.1 (Conjugate Matrix). Let $A = (a_{ij})_{m \times n} \in M_{m \times n}(\mathbb{C})$. The **conjugate matrix** $\bar{A}$ is obtained by taking the complex conjugate of each entry of A :

$$\bar{A} = (\bar{a}_{ij})_{m \times n}, \quad \text{where if } a_{ij} = x + iy, \text{ then } \bar{a}_{ij} = x - iy.$$

Definition 1.5.2 (Conjugate Transpose). Let $A \in M_{m \times n}(\mathbb{C})$. The **conjugate transpose** (also known as the **Hermitian transpose** or **adjoint matrix** in operator theory) of A , denoted by A^* (or A^H), is defined as the transpose of the complex conjugate of A : $A^* = (\bar{A})^T = \overline{(A^T)}$. Explicitly, if $A = (a_{ij})_{m \times n}$, then A^* is an $n \times m$ matrix whose (j, i) -th entry is:

$$(A^*)_{ji} = \bar{a}_{ij} \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

1.5.1 Algebraic Properties of Conjugate Transpose

Theorem 1.5.3 (Properties of Conjugate Transposition). *Let A, B be complex matrices of suitable sizes, and let $\alpha \in \mathbb{C}$. Then:*

- (i) **Involution Law:** $(A^*)^* = A$.
- (ii) **Sum Rule:** $(A + B)^* = A^* + B^*$.
- (iii) **Conjugate-Linearity:** $(\alpha A)^* = \bar{\alpha} A^*$. (Note the complex conjugation on the scalar α !)
- (iv) **Reversal Law for Multiplication:** $(AB)^* = B^* A^*$.
- (v) **Commutativity with Inversion:** If $A \in M_n(\mathbb{C})$ is invertible, then A^* is invertible and $(A^*)^{-1} = (A^{-1})^*$.

Proof. We prove properties (iii), (iv), and (v):

(iii) Let $A = (a_{ij})$. The (j, i) -th entry of $(\alpha A)^*$ is:

$$((\alpha A)^*)_{ji} = \overline{(\alpha a_{ij})} = \bar{\alpha} \bar{a}_{ij} = \bar{\alpha} (A^*)_{ji} = (\bar{\alpha} A^*)_{ji}.$$

Hence, $(\alpha A)^* = \bar{\alpha} A^*$.

(iv) Using the properties of transposition and complex conjugation:

$$(AB)^* = \overline{(AB)^T} = \overline{B^T A^T} = \overline{B^T} \cdot \overline{A^T} = B^* A^*.$$

(v) Taking the conjugate transpose of $AA^{-1} = I_n$ and using the reversal law:

$$(AA^{-1})^* = (I_n)^* \implies (A^{-1})^* A^* = I_n.$$

Similarly, $(A^{-1}A)^* = I_n^* \implies A^* (A^{-1})^* = I_n$.

By uniqueness of inverse, $(A^*)^{-1} = (A^{-1})^*$. ■

Remark 1.5.4. If A is a real matrix ($A \in M_{m \times n}(\mathbb{R})$), then $\bar{A} = A$. Consequently, for real matrices, the conjugate transpose reduces directly to the transpose:

$$A^* = A^T \quad \text{for all } A \in M_{m \times n}(\mathbb{R}).$$

Example 1.5.5. Consider the complex matrix $A = \begin{bmatrix} 1+i & 2-3i \\ 4 & 5i \end{bmatrix}$.

1. The conjugate of A is $\bar{A}$:

$$\bar{A} = \begin{bmatrix} 1-i & 2+3i \\ 4 & -5i \end{bmatrix}.$$

2. The transpose of $\bar{A}$ to find A^* :

$$A^* = (\bar{A})^T = \begin{bmatrix} 1-i & 4 \\ 2+3i & -5i \end{bmatrix}.$$

3. Therefore $(\alpha A)^* = \bar{\alpha} A^*$ for $\alpha = i$:

$$\alpha A = i \begin{bmatrix} 1+i & 2-3i \\ 4 & 5i \end{bmatrix} = \begin{bmatrix} -1+i & 3+2i \\ 4i & -5 \end{bmatrix}.$$

Taking conjugate transpose:

$$(\alpha A)^* = \begin{bmatrix} -1-i & -4i \\ 3-2i & -5 \end{bmatrix}.$$

On the other hand, $\bar{\alpha} = -i$:

$$\bar{\alpha}A^* = -i \begin{bmatrix} 1-i & 4 \\ 2+3i & -5i \end{bmatrix} = \begin{bmatrix} -i(1-i) & -4i \\ -i(2+3i) & 5i^2 \end{bmatrix} = \begin{bmatrix} -1-i & -4i \\ 3-2i & -5 \end{bmatrix}.$$

Thus $(\alpha A)^* = \bar{\alpha}A^*$ holds precisely.

1.6 Properties of Symmetric, Skew-Symmetric, Hermitian, and Skew-Hermitian Matrices

In matrix theory, square matrices with specific symmetries under transposition or conjugate transposition play a central role in linear algebra, physics, and engineering.

1.6.1 Symmetric and Skew-Symmetric Matrices

Definition 1.6.1 (Symmetric and Skew-Symmetric Matrices). Let $A = (a_{ij}) \in M_n(\mathbb{K})$ be a square matrix.

(i) A is called **symmetric** if $A^T = A$, which means:

$$a_{ij} = a_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

(ii) A is called **skew-symmetric** (or **anti-symmetric**) if $A^T = -A$, which means:

$$a_{ij} = -a_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

Theorem 1.6.2. All main diagonal entries of a skew-symmetric matrix are zero.

Proof. Let $A = (a_{ij}) \in M_n(\mathbb{K})$ be skew-symmetric. By definition, $a_{ij} = -a_{ji}$ for all i, j . For the diagonal entries ($i = j$), we have:

$$a_{ii} = -a_{ii} \implies 2a_{ii} = 0 \implies a_{ii} = 0.$$

Thus, every entry on the main diagonal of A must be zero. ■

Theorem 1.6.3. Let $A, B \in M_n(\mathbb{K})$ be symmetric matrices. The product AB is symmetric if and only if A and B commute ($AB = BA$).

Proof. Since A and B are symmetric, $A^T = A$ and $B^T = B$.

- ($\implies$) Suppose AB is symmetric. Then $(AB)^T = AB$. By the reversal law: $(AB)^T = B^T A^T = BA$. Therefore, $AB = BA$.
- ($\impliedby$) Suppose $AB = BA$. Then: $(AB)^T = B^T A^T = BA = AB$. Thus, AB is symmetric. ■

Theorem 1.6.4 (Symmetric Products AA^T and $A^T A$). For any rectangular matrix $A \in M_{m \times n}(\mathbb{K})$, the products AA^T (of order $m \times m$) and $A^T A$ (of order $n \times n$) are always symmetric.

Proof. Using the reversal law for transposition: $(AA^T)^T = (A^T)^T A^T = AA^T$, and $(A^T A)^T = A^T (A^T)^T = A^T A$. Hence, both AA^T and $A^T A$ are symmetric. ■

Theorem 1.6.5 (Determinant of Odd-Order Skew-Symmetric Matrices). *If $A \in M_n(\mathbb{K})$ is a skew-symmetric matrix and n is an odd integer, then $\det(A) = 0$.*

Proof. Since $A^T = -A$, taking the determinant on both sides yields:

$$\det(A^T) = \det(-A).$$

Using the property $\det(A^T) = \det(A)$ and $\det(cA) = c^n \det(A)$ for an $n \times n$ matrix:

$$\det(A) = (-1)^n \det(A).$$

Since n is odd, $(-1)^n = -1$, so:

$$\det(A) = -\det(A) \implies 2\det(A) = 0 \implies \det(A) = 0.$$

■

Theorem 1.6.6 (Decomposition into Symmetric and Skew-Symmetric Parts). *Every square matrix $A \in M_n(\mathbb{K})$ can be uniquely expressed as the sum of a symmetric matrix S and a skew-symmetric matrix K .*

Proof. Existence: Define S and K as follows:

$$S = \frac{1}{2}(A + A^T) \quad \text{and} \quad K = \frac{1}{2}(A - A^T).$$

First, we verify their structural properties:

$$S^T = \left(\frac{1}{2}(A + A^T) \right)^T = \frac{1}{2}(A^T + (A^T)^T) = \frac{1}{2}(A^T + A) = S \quad (S \text{ is symmetric}),$$

$$K^T = \left(\frac{1}{2}(A - A^T) \right)^T = \frac{1}{2}(A^T - (A^T)^T) = \frac{1}{2}(A^T - A) = -K \quad (K \text{ is skew-symmetric}).$$

Adding S and K :

$$S + K = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = A.$$

Uniqueness: Suppose $A = S_1 + K_1$, where $S_1^T = S_1$ and $K_1^T = -K_1$. Taking the transpose: $A^T = (S_1 + K_1)^T = S_1^T + K_1^T = S_1 - K_1$. Adding A and A^T : $A + A^T = 2S_1 \implies S_1 = \frac{1}{2}(A + A^T) = S$.

Subtracting A^T from A : $A - A^T = 2K_1 \implies K_1 = \frac{1}{2}(A - A^T) = K$.

Thus, the decomposition is unique. ■

Example 1.6.7. Decompose $A = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$ into symmetric and skew-symmetric parts:

$$A^T = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}.$$

$$S = \frac{1}{2}(A + A^T) = \frac{1}{2} \begin{bmatrix} 4 & 6 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix},$$

$$K = \frac{1}{2}(A - A^T) = \frac{1}{2} \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.$$

Clearly, $A = S + K$, with $S^T = S$ and $K^T = -K$.

1.6.2 Hermitian and Skew-Hermitian Matrices

Over the complex field $\mathbb{C}$, the complex conjugate transpose A^* plays the standard role of transpose.

Definition 1.6.8 (Hermitian and Skew-Hermitian Matrices). Let $A = (a_{ij}) \in M_n(\mathbb{C})$ be a complex square matrix.

(i) A is called **Hermitian** if $A^* = A$, which means:

$$a_{ij} = \bar{a}_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

(ii) A is called **skew-Hermitian** if $A^* = -A$, which means:

$$a_{ij} = -\bar{a}_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

Theorem 1.6.9. Let $A = (a_{ij}) \in M_n(\mathbb{C})$.

(i) If A is Hermitian, all its main diagonal entries are real.

(ii) If A is skew-Hermitian, all its main diagonal entries are either zero or purely imaginary.

Proof. (i) For a Hermitian matrix A , $a_{ij} = \bar{a}_{ji}$. For diagonal entries ($i = j$): $a_{ii} = \bar{a}_{ii}$. A complex number equal to its own conjugate must have an imaginary part of zero. Thus, $a_{ii} \in \mathbb{R}$.

(ii) For a skew-Hermitian matrix A , $a_{ij} = -\bar{a}_{ji}$. For diagonal entries ($i = j$):

$$a_{ii} = -\bar{a}_{ii} \implies a_{ii} + \bar{a}_{ii} = 0 \implies 2 \operatorname{Re}(a_{ii}) = 0 \implies \operatorname{Re}(a_{ii}) = 0.$$

Thus, a_{ii} is purely imaginary or zero. ■

Theorem 1.6.10. Let $A \in M_n(\mathbb{C})$.

(i) A is Hermitian if and only if iA is skew-Hermitian.

(ii) A is skew-Hermitian if and only if iA is Hermitian.

Proof. Using property $(\alpha M)^* = \bar{\alpha} M^*$:

(i) $(iA)^* = \bar{i} A^* = -i A^*$. If $A^* = A$ (Hermitian), then $(iA)^* = -iA = -(iA)$ (skew-Hermitian).

(ii) If $A^* = -A$ (skew-Hermitian), then $(iA)^* = -i(-A) = iA$ (Hermitian). ■

Theorem 1.6.11 (Hermitian Products AA^* and A^*A). For any rectangular complex matrix $A \in M_{m \times n}(\mathbb{C})$, AA^* and A^*A are always Hermitian.

Proof. By the reversal law for conjugate transpose:

$$(AA^*)^* = (A^*)^* A^* = AA^*, \quad \text{and} \quad (A^*A)^* = A^*(A^*)^* = A^*A.$$

Hence, AA^* and A^*A are Hermitian. ■

Theorem 1.6.12 (Decomposition into Hermitian and Skew-Hermitian Parts). Every square complex matrix $A \in M_n(\mathbb{C})$ can be uniquely decomposed as: $A = H + K$, where H is Hermitian ($H^* = H$) and K is skew-Hermitian ($K^* = -K$).

Proof. Construct $H = \frac{1}{2}(A + A^*)$ and $K = \frac{1}{2}(A - A^*)$. Checking conjugate transposes:

$$H^* = \left(\frac{1}{2}(A + A^*)\right)^* = \frac{1}{2}(A^* + A) = H \quad (\text{Hermitian}),$$

$$K^* = \left(\frac{1}{2}(A - A^*)\right)^* = \frac{1}{2}(A^* - A) = -K \quad (\text{skew-Hermitian}).$$

Summing: $H + K = A$. Uniqueness follows identically to the symmetric case using A^* . ■

Example 1.6.13. Decompose $A = \begin{bmatrix} 2+i & 3 \\ 1-i & 4i \end{bmatrix}$ into Hermitian and skew-Hermitian components:

$$A^* = (\bar{A})^T = \begin{bmatrix} 2-i & 1+i \\ 3 & -4i \end{bmatrix}.$$

$$H = \frac{1}{2}(A + A^*) = \frac{1}{2} \begin{bmatrix} 4 & 4+i \\ 4-i & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2+\frac{1}{2}i \\ 2-\frac{1}{2}i & 0 \end{bmatrix},$$

$$K = \frac{1}{2}(A - A^*) = \frac{1}{2} \begin{bmatrix} 2i & 2-i \\ -2-i & 8i \end{bmatrix} = \begin{bmatrix} i & 1-\frac{1}{2}i \\ -1-\frac{1}{2}i & 4i \end{bmatrix}.$$

Notice $H^* = H$ (diagonal entries are real: 2, 0) and $K^* = -K$ (diagonal entries are purely imaginary: $i, 4i$).

Theorem 1.6.14 (Cartesian Decomposition of a Matrix). *Every square complex matrix $A \in M_n(\mathbb{C})$ can be uniquely expressed in the form:*

$$A = P + iQ,$$

where both P and Q are Hermitian matrices (i.e., $P^* = P$ and $Q^* = Q$).

Proof. Existence. Let $A \in M_n(\mathbb{C})$ be any square complex matrix, and let A^* be its conjugate transpose. Define two matrices P and Q as:

$$P = \frac{1}{2}(A + A^*) \quad \text{and} \quad Q = \frac{1}{2i}(A - A^*).$$

We first verify that both P and Q are Hermitian:

We have

$$P^* = \left(\frac{1}{2}(A + A^*)\right)^* = \frac{1}{2}(A^* + (A^*)^*) = \frac{1}{2}(A^* + A) = P.$$

Thus, P is Hermitian ($P^* = P$).

Also,

$$Q^* = \left(\frac{1}{2i}(A - A^*)\right)^* = -\frac{1}{2i}(A^* - (A^*)^*) = -\frac{1}{2i}(A^* - A) = \frac{1}{2i}(A - A^*) = Q.$$

Thus, Q is also Hermitian ($Q^* = Q$).

Therefore,

$$P + iQ = \frac{1}{2}(A + A^*) + i\left(\frac{1}{2i}(A - A^*)\right) = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) = A.$$

This proves that the decomposition $A = P + iQ$ exists.

Uniqueness. Suppose there exist two Hermitian matrices R and S (so $R^* = R$ and $S^* = S$) such that: $A = R + iS$. Taking the conjugate transpose of both sides:

$$A^* = (R + iS)^* = R^* + (iS)^* = R^* - iS^* = R - iS.$$

We now have a system of two linear matrix equations:

$$A = R + iS \quad (1)$$

$$A^* = R - iS \quad (2)$$

Adding equations (1) and (2):

$$A + A^* = 2R \implies R = \frac{1}{2}(A + A^*) = P.$$

Subtracting equation (2) from equation (1):

$$A - A^* = 2iS \implies S = \frac{1}{2i}(A - A^*) = Q.$$

Since $R = P$ and $S = Q$, the decomposition $A = P + iQ$ is unique. ■

Remark 1.6.15 (Connection to Hermitian and Skew-Hermitian Decomposition). The standard Cartesian decomposition is $A = P + iQ$, where both P and Q are Hermitian. Since iQ is skew-Hermitian, this is equivalent to writing $A = P + K$, where P is Hermitian and K is skew-Hermitian. Indeed, if we set $K = iQ$, then $K^* = (iQ)^* = -iQ^* = -iQ = -K$. Thus, $K = iQ$ is a **skew-Hermitian** matrix. Hence, the Cartesian decomposition $A = P + iQ$ (where P, Q are Hermitian) is mathematically equivalent to writing $A = P + K$ (where P is Hermitian and K is skew-Hermitian).

Example 1.6.16. Decompose the matrix $A = \begin{bmatrix} 2+3i & 1-i \\ 3+5i & 4 \end{bmatrix}$ into the form $P + iQ$, where P and Q are Hermitian.

Solution: First, compute the conjugate transpose $A^* = (\bar{A})^T$:

$$\bar{A} = \begin{bmatrix} 2-3i & 1+i \\ 3-5i & 4 \end{bmatrix} \implies A^* = \begin{bmatrix} 2-3i & 3-5i \\ 1+i & 4 \end{bmatrix}.$$

1. Calculate $P = \frac{1}{2}(A + A^*)$:

$$A + A^* = \begin{bmatrix} (2+3i) + (2-3i) & (1-i) + (3-5i) \\ (3+5i) + (1+i) & 4+4 \end{bmatrix} = \begin{bmatrix} 4 & 4-6i \\ 4+6i & 8 \end{bmatrix}.$$

$$P = \frac{1}{2} \begin{bmatrix} 4 & 4-6i \\ 4+6i & 8 \end{bmatrix} = \begin{bmatrix} 2 & 2-3i \\ 2+3i & 4 \end{bmatrix}.$$

Notice $P^* = P$ (diagonal entries are real: 2, 4).

2. Calculate $Q = \frac{1}{2i}(A - A^*)$:

$$A - A^* = \begin{bmatrix} (2+3i) - (2-3i) & (1-i) - (3-5i) \\ (3+5i) - (1+i) & 4-4 \end{bmatrix} = \begin{bmatrix} 6i & -2+4i \\ 2+4i & 0 \end{bmatrix}.$$

Dividing by $2i$ (note that $\frac{1}{2i} = -\frac{i}{2}$):

$$Q = \frac{1}{2i} \begin{bmatrix} 6i & -2+4i \\ 2+4i & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2+i \\ 2-i & 0 \end{bmatrix}.$$

Thus, $Q^* = Q$.

3. Verification:

$$P + iQ = \begin{bmatrix} 2 & 2-3i \\ 2+3i & 4 \end{bmatrix} + i \begin{bmatrix} 3 & 2+i \\ 2-i & 0 \end{bmatrix} = \begin{bmatrix} 2+3i & (2-3i) + (2i-1) \\ (2+3i) + (2i+1) & 4+0 \end{bmatrix}$$

$$= \begin{bmatrix} 2+3i & 1-i \\ 3+5i & 4 \end{bmatrix} = A.$$

This completes the decomposition.

Matrix Type	Defining Condition	Diagonal Entries	Real Analogue
Symmetric	$A^T = A$	Unrestricted	$a_{ij} = a_{ji}$
Skew-Symmetric	$A^T = -A$	All Zero ($a_{ii} = 0$)	$a_{ij} = -a_{ji}$
Hermitian	$A^* = A$	Purely Real ($\text{Im}(a_{ii}) = 0$)	Symmetric
Skew-Hermitian	$A^* = -A$	Purely Imaginary or Zero ($\text{Re}(a_{ii}) = 0$)	Skew-Symmetric

Summary comparison.

Exercise 1.6.17. Solve the following exercises:

- Construct two 2×2 real matrices A and B such that $AB = 0$ (the zero matrix) but $BA \neq 0$. What does this imply about zero divisors in matrix algebra?
- Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$. Compute $(A + B)^2$ and $A^2 + 2AB + B^2$. Show that $(A + B)^2 \neq A^2 + 2AB + B^2$, and state the necessary and sufficient condition under which equality holds.
- Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ be matrices over $\mathbb{C}$. Compute AB and BA , and show that A and B anti-commute (i.e., $AB = -BA$).
- Verify the reversal law for transposition $(AB)^T = B^T A^T$ for the matrices:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ -1 & 1 \end{bmatrix}.$$

- Using the reversal law for two matrices, prove by induction that for any k conformable matrices $A_1, A_2, \dots, A_k$:

$$(A_1 A_2 \cdots A_k)^T = A_k^T \cdots A_2^T A_1^T.$$

- For $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, find A^{-1} and explicitly verify that $AA^{-1} = A^{-1}A = I_2$.
- Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, calculate $(AB)^{-1}$ and $B^{-1}A^{-1}$, and confirm that $(AB)^{-1} = B^{-1}A^{-1}$.
- Show that if $A \in M_n(\mathbb{K})$ is an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$. Verify this result for $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$.
- Let $A = \begin{bmatrix} 2+i & 3-2i \\ 4i & 1+5i \end{bmatrix} \in M_2(\mathbb{C})$. Find the conjugate transpose A^* , and verify that $(2iA)^* = -2iA^*$.
- Let $A \in M_{m \times p}(\mathbb{C})$ and $B \in M_{p \times n}(\mathbb{C})$. Prove formally from definition that $(AB)^* = B^*A^*$.
- Express the real matrix $A = \begin{bmatrix} 3 & 8 & 2 \\ 2 & 5 & 6 \\ 4 & 0 & 1 \end{bmatrix}$ uniquely in the form $A = S + K$, where S is symmetric ($S^T = S$) and K is skew-symmetric ($K^T = -K$).

12. Prove that if A is an $n \times n$ skew-symmetric matrix and n is an odd integer, then $\det(A) = 0$. Using this property, state $\det(A)$ without expansion for:

$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}.$$

13. Prove that all main diagonal entries of a Hermitian matrix are real numbers, whereas all main diagonal entries of a skew-Hermitian matrix are either zero or purely imaginary.
14. Show that if H is a Hermitian matrix, then $K = iH$ is a skew-Hermitian matrix. Conversely, show that if K is skew-Hermitian, then $H = -iK$ is Hermitian.
15. Decompose the complex matrix $A = \begin{bmatrix} 1+2i & 3-i \\ 2+4i & 5i \end{bmatrix}$ into the form $A = P + iQ$, where P is a Hermitian matrix and Q is a skew-Hermitian matrix.
16. Find the cofactor matrix C and the adjoint matrix $\text{adj}(A)$ for:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}.$$

17. For $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}$, compute $\text{adj}(A)$ and verify the fundamental identity $A \cdot \text{adj}(A) = \det(A)I_3$.

18. Let A be a 3×3 non-singular matrix with $\det(A) = 4$.

- (a) Prove that $\det(\text{adj}(A)) = (\det(A))^2$.
- (b) Find the numerical value of $\det(\text{adj}(A))$.

19. Prove that for any non-singular matrix $A \in M_n(\mathbb{K})$ ($n \geq 2$):

$$\text{adj}(\text{adj}(A)) = (\det(A))^{n-2}A.$$

20. Using the adjoint method ($A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$), determine the inverse of the matrix:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}.$$

Chapter 2

Elementary Operations and Systems of Linear Equations

This chapter develops the use of elementary matrix operations in row reduction, rank, normal forms, invertibility, and factorization by elementary matrices. These ideas provide the algebraic tools needed for the systematic study of linear systems.

2.1 Elementary Operations, Row Reduction, Normal Form, and Elementary Matrices

Elementary row operations simplify a matrix without changing its essential row-equivalence properties. They are used to determine rank, test invertibility, compute inverses, and reduce matrices to standard forms.

Definition 2.1.1 (Elementary Row Operations). Let A be an $m \times n$ matrix over a field $\mathbb{K}$. The following operations are called **elementary row operations**:

Row interchange: $R_i \leftrightarrow R_j$.

Row scaling: $R_i \rightarrow cR_i$, where $c \in \mathbb{K}$ and $c \neq 0$.

Row replacement: $R_i \rightarrow R_i + cR_j$, where $i \neq j$.

Each elementary row operation is reversible. The inverse of $R_i \leftrightarrow R_j$ is the same interchange, the inverse of $R_i \rightarrow cR_i$ is $R_i \rightarrow c^{-1}R_i$, and the inverse of $R_i \rightarrow R_i + cR_j$ is $R_i \rightarrow R_i - cR_j$.

Definition 2.1.2 (Elementary Matrix). An **elementary matrix** is a square matrix obtained by performing one elementary row operation on an identity matrix.

For example, applying $R_2 \rightarrow R_2 - 3R_1$ to I_3 gives $E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Theorem 2.1.3 (Representation of Elementary Row Operations). Let A be an $m \times n$ matrix and let E be obtained by applying an elementary row operation e to I_m . Then the same operation on A is represented by left multiplication: $e(A) = EA$.

Proof. The rows of EA are formed from the rows of A in exactly the same way that the rows of E are formed from the rows of I_m . Thus row interchange, row scaling, and row replacement on I_m produce the corresponding operations on A . ■

Theorem 2.1.4 (Representation of Elementary Column Operations). *Let A be an $m \times n$ matrix and let F be the elementary matrix corresponding to an elementary column operation f . Then $f(A) = AF$.*

Proof. A column operation on A is the transpose of the corresponding row operation on A^T . If E represents that row operation, then $e(A^T) = EA^T$, and therefore $f(A) = (EA^T)^T = AE^T$. Taking $F = E^T$ gives the required result. ■

Remark 2.1.5. Elementary row operations are represented by left multiplication, while elementary column operations are represented by right multiplication. Thus $A \rightarrow EA$ changes rows and $A \rightarrow AF$ changes columns.

Theorem 2.1.6 (Invertibility of Elementary Matrices). *Every elementary matrix is non-singular, and its inverse is also an elementary matrix.*

Proof. The inverse of each elementary operation is again elementary. Hence, if E' represents the inverse operation of the one represented by E , then $EE' = E'E = I$. Therefore $E^{-1} = E'$. ■

Example 2.1.7. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$. Construct elementary matrices for the operations $R_1 \leftrightarrow R_3$, $R_2 \rightarrow 4R_2$, and $R_3 \rightarrow R_3 - 2R_1$.

Solution. Applying the operations to I_3 gives

$$E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

Thus E_1A interchanges rows 1 and 3, E_2A multiplies row 2 by 4, and E_3A replaces row 3 by $R_3 - 2R_1$. ■

Example 2.1.8. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Represent the column operation $C_2 \rightarrow C_2 - 2C_1$ by an elementary matrix.

Solution. Applying the same column operation to I_2 gives $F = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$. Hence

$$AF = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix},$$

which is exactly the result of replacing C_2 by $C_2 - 2C_1$. ■

Example 2.1.9. For $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, find E^{-1} .

Solution. The matrix E represents $R_2 \rightarrow 4R_2$. The inverse operation is $R_2 \rightarrow \frac{1}{4}R_2$, so

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Definition 2.1.10 (Row Echelon Form). A matrix is in **Row Echelon Form (REF)** if:

- (i) all zero rows occur below the non-zero rows;
- (ii) the first non-zero entry of each non-zero row, called its **pivot**, lies strictly to the right of the pivot in the row above it.

Definition 2.1.11 (Reduced Row Echelon Form). A matrix is in **Reduced Row Echelon Form (RREF)** if it is in REF and, in addition, every pivot is 1 and is the only non-zero entry in its column.

Theorem 2.1.12 (Uniqueness of RREF). *Every matrix is row equivalent to a unique matrix in Reduced Row Echelon Form.*

Proof. Existence follows from the row-reduction algorithm. For uniqueness, suppose R and S are RREF matrices row equivalent to the same matrix. Then the homogeneous systems $RX = 0$ and $SX = 0$ have the same solution set. This forces the same pivot columns in R and S . Once the pivot columns are fixed, the basic solutions obtained from the free variables determine every non-pivot column uniquely. Hence $R = S$. ■

Remark 2.1.13. REF is generally not unique, whereas RREF is unique. Elementary row operations preserve rank, and the rank of a matrix equals the number of pivots in any row echelon form.

Definition 2.1.14 (Row Equivalent Matrices). Two matrices of the same size are **row equivalent** if one can be obtained from the other by a finite sequence of elementary row operations.

Example 2.1.15. Reduce $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 8 & 7 \end{bmatrix}$ to REF and determine its rank.

Solution. Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$, followed by $R_3 \rightarrow R_3 - 2R_2$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 8 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The final matrix is in REF and has two pivots. Hence $\text{rank}(A) = 2$. ■

Example 2.1.16. Reduce $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 1 & 2 & 3 \end{bmatrix}$ to RREF.

Solution. Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - R_1$, then $R_3 \rightarrow R_3 - R_2$ and $R_1 \rightarrow R_1 - R_2$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 1 & 2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $\text{RREF}(A) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $\text{rank}(A) = 2$. ■

Definition 2.1.17 (Normal Form of a Matrix). If A is an $m \times n$ matrix of rank r , then

$$N = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}$$

is called the **normal form** or **rank canonical form** of A .

Remark 2.1.18. RREF is obtained using row operations only. Normal form is obtained using both row and column operations. An RREF matrix may contain non-zero entries in non-pivot columns, while the normal form has zeros outside the identity block.

Theorem 2.1.19 (Reduction to Normal Form). *If A is an $m \times n$ matrix of rank r , then there exist non-singular matrices $P \in M_m(\mathbb{K})$ and $Q \in M_n(\mathbb{K})$ such that*

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

Proof. Row reduction gives a non-singular matrix P such that $PA = R$, where R has exactly r pivots. Elementary column operations then move the pivot columns to the first r positions and clear the remaining entries in the pivot rows. Their product is a non-singular matrix Q , giving the stated normal form. ■

Remark 2.1.20 (Computation of P and Q). Write $A = I_m A I_n$. Apply each row operation on A also to I_m ; the resulting matrix is P . After the row reduction, apply each column operation on the reduced matrix also to I_n ; the resulting matrix is Q .

Example 2.1.21. The matrix $R = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$ is in RREF but not in normal form. Reduce it to normal form.

Solution. The first two columns are pivot columns. Applying $C_3 \rightarrow C_3 - 2C_1 - 3C_2$ gives

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus the normal form is $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$. ■

Example 2.1.22. Find non-singular matrices P and Q such that PAQ is in normal form for

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix}.$$

Solution. Record the row operations on I_3 . Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 6 \end{bmatrix}, \quad I_3 \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

Now apply $R_3 \rightarrow R_3 - 2R_2$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}.$$

Hence $P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}$ and $PA = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$.

Next record the column operations on I_3 . Apply $C_2 \rightarrow C_2 - C_1$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}, \quad I_3 \longrightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Apply $C_3 \rightarrow C_3 - 2C_1$:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Finally apply $C_3 \rightarrow C_3 - 3C_2$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

and the same operation gives $Q = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$. Therefore

$$P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix},$$

and $PAQ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Hence $\text{rank}(A) = 2$. ■

Definition 2.1.23 (Non-Singular Matrix). A square matrix A is **non-singular** or **invertible** if there exists a matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$. Otherwise, A is **singular**.

Theorem 2.1.24 (Invertibility and Row Reduction). For an $n \times n$ matrix A , the following statements are equivalent:

- (i) A is invertible;
- (ii) A is row equivalent to I_n ;
- (iii) $\text{RREF}(A) = I_n$;
- (iv) A has n pivots;
- (v) $\text{rank}(A) = n$;
- (vi) the homogeneous system $AX = 0$ has only the trivial solution.

Proof. If A is row equivalent to I_n , then for some elementary matrices $E_1, \dots, E_k$, $E_k E_{k-1} \cdots E_1 A = I_n$. Hence $A^{-1} = E_k E_{k-1} \cdots E_1$, so A is invertible. Conversely, if A is invertible, then $\text{rank}(A) = n$, so its RREF has n pivots and must be I_n . The remaining equivalences follow from the relation between rank, pivots, and the solution set of $AX = 0$. ■

Corollary 2.1.25. An $n \times n$ matrix is singular if and only if its RREF has fewer than n pivots.

Theorem 2.1.26 (Finding the Inverse by Row Reduction). If A is invertible, then elementary row operations reduce the augmented matrix according to $[A \mid I_n] \longrightarrow [I_n \mid A^{-1}]$.

Proof. If $E_k \cdots E_1 A = I_n$, then $E_k \cdots E_1 = A^{-1}$. Applying the same row operations to I_n therefore produces A^{-1} . ■

Example 2.1.27. Determine whether $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ is invertible.

Solution. Apply $R_2 \rightarrow R_2 - 3R_1$, $R_2 \rightarrow -R_2$, and $R_1 \rightarrow R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \sim I_2.$$

Thus $\text{RREF}(A) = I_2$, so A is invertible. ■

Example 2.1.28. Determine whether $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ is invertible.

Solution. Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}.$$

The matrix has only two pivots, so $\text{rank}(A) = 2 < 3$. Hence A is singular. ■

Example 2.1.29. Find the inverse of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by row reduction.

Solution. Begin with

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right].$$

Apply $R_2 \rightarrow R_2 - 3R_1$, $R_2 \rightarrow -\frac{1}{2}R_2$, and $R_1 \rightarrow R_1 - 2R_2$:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right].$$

Therefore

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.$$

Theorem 2.1.30 (Non-Singular Matrices as Products of Elementary Matrices). *Every non-singular square matrix can be expressed as a finite product of elementary matrices.*

Proof. If A is non-singular, then $\text{RREF}(A) = I_n$. Hence there exist elementary matrices $E_1, \dots, E_k$ such that $E_k \cdots E_1 A = I_n$. Therefore $A = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$. Since the inverse of an elementary matrix is elementary, A is a product of elementary matrices. ■

Corollary 2.1.31. *A square matrix is non-singular if and only if it is a finite product of elementary matrices.*

Proof. The forward implication follows from the theorem. Conversely, every elementary matrix is invertible, and a product of invertible matrices is invertible. ■

Remark 2.1.32. If $E_k \cdots E_1 A = I$, then $A^{-1} = E_k \cdots E_1$, while $A = E_1^{-1} \cdots E_k^{-1}$. Thus the same row reduction gives both the inverse of A and a factorization of A into elementary matrices.

Example 2.1.33. Express $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ as a product of elementary matrices.

Solution. Reduce A to I_2 using $R_2 \rightarrow \frac{1}{3}R_2$ and $R_1 \rightarrow R_1 - 2R_2$. The corresponding elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Since $E_2E_1A = I_2$, we have $A = E_1^{-1}E_2^{-1}$. Hence

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

■

Example 2.1.34. Express $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ as a product of elementary matrices.

Solution. Reduce A to I_3 using $R_3 \rightarrow \frac{1}{3}R_3$, $R_2 \rightarrow R_2 - 2R_3$, and $R_1 \rightarrow R_1 - R_2$. The corresponding elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since $E_3E_2E_1A = I_3$, it follows that

$$A = E_1^{-1}E_2^{-1}E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

■

2.2 Systems of Linear Equations and Their Solution Sets

A system of m linear equations in n unknowns can be written in matrix form as $AX = B$, where $A \in M_{m \times n}(\mathbb{K})$ is the coefficient matrix, $X = (x_1, \dots, x_n)^T \in \mathbb{K}^n$ is the vector of unknowns, and $B \in \mathbb{K}^m$ is the right-hand side. Row reduction provides a systematic method for deciding whether the system is consistent and, when it is consistent, for describing all its solutions.

Explicitly, if $A = (a_{ij})_{m \times n}$, $X = (x_1, \dots, x_n)^T$, and $B = (b_1, \dots, b_m)^T$, then $AX = B$ represents

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

Definition 2.2.1 (Homogeneous and Non-Homogeneous Systems). The system $AX = B$ is called **homogeneous** when $B = 0$ and **non-homogeneous** when $B \neq 0$. For a homogeneous system, $X = 0$ is called the **trivial solution**; any solution $X \neq 0$ is called a **non-trivial solution**.

Remark 2.2.2. Every homogeneous system is consistent because $X = 0$ always satisfies $AX = 0$. A non-homogeneous system may have no solution, exactly one solution, or infinitely many solutions.

Remark 2.2.3. For a homogeneous system $AX = 0$, the complete set of solutions is obtained by reducing A to RREF and expressing the pivot variables in terms of the free variables.

Theorem 2.2.4 (Existence of Non-Trivial Solutions). *Let A be an $m \times n$ matrix of rank r . The homogeneous system $AX = 0$ has a non-trivial solution if and only if $r < n$. Equivalently, $AX = 0$ has only the trivial solution if and only if $\text{rank}(A) = n$.*

Proof. After reducing A to RREF, there are r pivot variables and $n - r$ free variables. If $r = n$, no free variable remains and the only solution is $X = 0$. If $r < n$, at least one free variable is present; assigning a non-zero value to a free variable gives a non-trivial solution. ■

Corollary 2.2.5. *If A is an $m \times n$ matrix with $n > m$, then $AX = 0$ has a non-trivial solution.*

Proof. Since $\text{rank}(A) \leq m < n$, we have $\text{rank}(A) < n$. ■

Remark 2.2.6 (Systematic Solution of $AX = 0$). To solve a homogeneous system, reduce A to RREF, identify pivot and free variables, express the pivot variables in terms of the free variables, and write the solution in parametric vector form.

Example 2.2.7. Solve $x + 2y - z = 0$, $2x + 5y + z = 0$, $3x + 7y = 0$.

Solution. The coefficient matrix reduces as

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 5 & 1 \\ 3 & 7 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $x - 7z = 0$ and $y + 3z = 0$. Taking $z = t$, we get $x = 7t$ and $y = -3t$. Hence

$$X = t \begin{bmatrix} 7 \\ -3 \\ 1 \end{bmatrix}, \quad t \in \mathbb{K}.$$

Thus all solutions are scalar multiples of $\begin{bmatrix} 7 \\ -3 \\ 1 \end{bmatrix}$. ■

Example 2.2.8. Solve $x + y + z + w = 0$, $2x + y + 3z + 2w = 0$.

Solution. Row reduction gives

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

Hence $x = -2z - w$ and $y = z$. Let $z = s$ and $w = t$. Then

$$X = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{K}.$$

Thus the complete solution contains two arbitrary parameters s and t . ■

Example 2.2.9. Solve $AX = 0$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

Solution. Since $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \sim I_2$, both variables are pivot variables and there are no free variables. Hence $X = 0$ is the only solution. ■

Example 2.2.10. Find the general solution of $x + 2y + z + 3w = 0$, $2x + 4y + 2z + 6w = 0$.

Solution. The second equation is twice the first, so the system reduces to $x + 2y + z + 3w = 0$. Let $y = s$, $z = t$, and $w = u$. Then $x = -2s - t - 3u$, and

$$X = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} -3 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Thus the complete solution depends on the three arbitrary parameters $s, t, u \in \mathbb{K}$. ■

Definition 2.2.11 (Consistency). A system $AX = B$ is called **consistent** if it has at least one solution and **inconsistent** if it has no solution.

For a non-homogeneous system, form the augmented matrix $[A \mid B]$ and reduce it by elementary row operations. A row of the form $[0 \cdots 0 \mid c]$ with $c \neq 0$ represents the impossible equation $0 = c$ and therefore shows that the system is inconsistent.

Theorem 2.2.12 (Rouché–Capelli Theorem). The system $AX = B$ is consistent if and only if

$$\text{rank}(A) = \text{rank}([A \mid B]).$$

If the common rank is r and A has n columns, then:

- (i) if $r = n$, the system has a unique solution;
- (ii) if $r < n$, the system has infinitely many solutions.

Proof. If $\text{rank}([A \mid B]) > \text{rank}(A)$, row reduction produces a contradictory row $[0 \cdots 0 \mid c]$ with $c \neq 0$, so the system is inconsistent. If the two ranks are equal, no contradictory row occurs and the system is consistent. If the common rank equals n , every variable is a pivot variable; if it is less than n , at least one free variable remains. ■

Remark 2.2.13. The rank condition has a simple interpretation: the system is consistent precisely when the right-hand side B can be obtained from the columns of A using suitable values of the unknowns. Equivalently, adding B as an extra column must not increase the rank.

Remark 2.2.14 (Consistency Criterion). For a system of n unknowns:

- (i) $\text{rank}(A) \neq \text{rank}([A \mid B]) \implies$ no solution;
- (ii) $\text{rank}(A) = \text{rank}([A \mid B]) = n \implies$ unique solution;
- (iii) $\text{rank}(A) = \text{rank}([A \mid B]) < n \implies$ infinitely many solutions.

Theorem 2.2.15. Let A be an $n \times n$ matrix. The system $AX = B$ has a unique solution for every $B \in \mathbb{K}^n$ if and only if A is non-singular.

Proof. If A is non-singular, then $X = A^{-1}B$ is the unique solution. Conversely, if $AX = B$ has a unique solution for every B , then in particular $AX = 0$ has only the trivial solution. Hence $\text{rank}(A) = n$, so A is non-singular. ■

Remark 2.2.16 (Systematic Solution of $AX = B$). Form $[A \mid B]$ and reduce it to REF or RREF. First check for a contradictory row. If none occurs, identify pivot and free variables and write the complete solution in terms of the free parameters.

Example 2.2.17. Solve $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$.

Solution. The augmented matrix reduces as

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array} \right].$$

Thus $z = 3$, $y = 2$, and $x = 1$. Hence $X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Since $\text{rank}(A) = \text{rank}([A \mid B]) = 3$, the solution is unique. ■

Example 2.2.18. Solve $x + y + z = 2$, $2x + 2y + 2z = 4$, $x - y + z = 0$.

Solution. The augmented matrix reduces to

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $x + z = 1$ and $y = 1$. Let $z = t$. Then

$$X = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad t \in \mathbb{K}.$$

Here $\text{rank}(A) = \text{rank}([A \mid B]) = 2 < 3$, so the system has infinitely many solutions. ■

Example 2.2.19. Show that $x + 2y - z = 1$, $2x + 4y - 2z = 3$ has no solution.

Solution. The augmented matrix reduces as

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 4 & -2 & 3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

The second row represents $0 = 1$. Therefore $\text{rank}(A) = 1$ and $\text{rank}([A \mid B]) = 2$, so the system is inconsistent. ■

Example 2.2.20. Determine the values of λ for which $x + y + z = 1$, $2x + 2y + 2z = \lambda$, $x - y + z = 0$ is consistent, and find the solution set when it is consistent.

Solution. Row reduction gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & \lambda \\ 1 & -1 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & \lambda - 2 \\ 0 & -2 & 0 & -1 \end{array} \right].$$

The system is consistent exactly when $\lambda - 2 = 0$, so $\lambda = 2$. For $\lambda = 2$, we get $y = \frac{1}{2}$ and $x + z = \frac{1}{2}$. Taking $z = t$,

$$X = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad t \in \mathbb{K}.$$

Hence the system has infinitely many solutions for $\lambda = 2$ and no solution for $\lambda \neq 2$. ■

Example 2.2.21. Determine the values of λ for which the system

$$x + y + z = 2, \quad x + \lambda y + z = 3, \quad 2x + (\lambda + 1)y + 2z = 5$$

has a unique solution, infinitely many solutions, or no solution.

Solution. The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & \lambda & 1 & 3 \\ 2 & \lambda + 1 & 2 & 5 \end{array} \right].$$

Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & \lambda - 1 & 0 & 1 \\ 0 & \lambda - 1 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & \lambda - 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

If $\lambda \neq 1$, the second row gives a pivot in the y -column, but the z -column remains free. Hence $\text{rank}(A) = \text{rank}([A \mid B]) = 2 < 3$, so the system has infinitely many solutions.

If $\lambda = 1$, the second row becomes $[0 \ 0 \ 0 \mid 1]$, which is inconsistent. Hence the system has no solution.

Therefore this system never has a unique solution: it has infinitely many solutions for $\lambda \neq 1$ and no solution for $\lambda = 1$. ■

Example 2.2.22. Determine the values of λ for which the system

$$x + y + z = 1, \quad x + \lambda y + 2z = 2, \quad x + 2y + \lambda z = 3$$

has a unique solution.

Solution. The coefficient matrix is $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & 2 \\ 1 & 2 & \lambda \end{bmatrix}$. A square system has a unique solution exactly when A is non-singular. Hence we require $\det A \neq 0$. Now

$$\det A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 2 \\ 1 & 2 & \lambda \end{vmatrix} = \lambda^2 - 2\lambda = (\lambda)(\lambda - 2).$$

Therefore the system has a unique solution when $\lambda \neq 0, 2$. For $\lambda = 0$ or $\lambda = 2$, row reduction of the augmented matrix is needed to decide whether the system has infinitely many solutions or no solution. ■

Example 2.2.23. For the system

$$x + y + z = 1, \quad x + \lambda y + 2z = 2, \quad x + 2y + \lambda z = 3,$$

determine what happens when $\lambda = 0$ and $\lambda = 2$.

Solution. For $\lambda = 0$,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{array} \right].$$

Thus $\lambda = 0$ gives no solution.

For $\lambda = 2$,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 2 & 3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

Thus $\lambda = 2$ also gives no solution. ■

Remark 2.2.24. When a parameter occurs in a square system, the determinant is useful for identifying values for which a unique solution is possible. For parameter values making the determinant zero, row reduction of the augmented matrix is still required to distinguish between infinitely many solutions and no solution.

Example 2.2.25. Find the values of k for which $x + y + z = 2$, $2x + 3y + 4z = 5$, $3x + 4y + kz = 7$ has a unique solution.

Solution. The coefficient matrix is $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & k \end{bmatrix}$. A unique solution exists when $\det A \neq 0$.

Expanding,

$$\det A = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & k \end{vmatrix} = k - 5.$$

Hence the system has a unique solution for $k \neq 5$. When $k = 5$, the coefficient matrix is singular and consistency must be tested separately. ■

Example 2.2.26. For $k = 5$, determine whether $x + y + z = 2$, $2x + 3y + 4z = 5$, $3x + 4y + 5z = 7$ has no solution or infinitely many solutions.

Solution. The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 7 \end{array} \right].$$

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\text{rank}(A) = \text{rank}([A \mid B]) = 2 < 3$. Hence the system has infinitely many solutions. ■

Theorem 2.2.27 (Relation Between Homogeneous and Non-Homogeneous Solutions). Suppose $AX = B$ is consistent and X_p is one particular solution. Then every solution of $AX = B$ is obtained by adding to X_p a solution X_h of the associated homogeneous system $AX = 0$: $X = X_p + X_h$.

Proof. If X and X_p both satisfy $AX = B$, then $A(X - X_p) = 0$. Hence $X - X_p$ is a solution of $AX = 0$, so $X = X_p + X_h$. Conversely, if $AX_h = 0$, then $A(X_p + X_h) = B$. ■

Remark 2.2.28. Therefore, to describe all solutions of a consistent non-homogeneous system, it is enough to find one particular solution and then add the general solution of the associated homogeneous system.

Corollary 2.2.29. If $AX = B$ is consistent, A has n columns, and $\text{rank}(A) = r$, then the complete solution contains $n - r$ arbitrary parameters.

Example 2.2.30. Describe the complete solution set of $x + y + z = 3$, $2x + 2y + 2z = 6$ as a particular solution plus the general solution of the associated homogeneous system.

Solution. The system reduces to $x + y + z = 3$. Choose the particular solution $X_p = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$. The associated homogeneous equation is $x + y + z = 0$, whose general solution is

$$X_h = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore

$$X = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{K}.$$

Example 2.2.31. For $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, compare the general solutions of $AX = B$ and $AX = 0$.

Solution. The equations $x + y = 2$ and $y + z = 3$ give, with $y = t$,

$$X = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}.$$

For $AX = 0$, the solution is

$$X_h = t \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}.$$

Thus the general solution of $AX = B$ is obtained by adding the particular solution $X_p = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ to the general solution of $AX = 0$.

Remark 2.2.32 (Summary). For $AX = 0$, non-trivial solutions exist exactly when $\text{rank}(A) < n$, where n is the number of unknowns. If $\text{rank}(A) = r$, then the general solution contains $n - r$ arbitrary parameters. For $AX = B$, consistency is characterized by $\text{rank}(A) = \text{rank}([A \mid B])$. A consistent system has a unique solution when the common rank equals the number of unknowns and infinitely many solutions when it is smaller. When parameters occur in the equations, exceptional parameter values should be examined separately by row reduction. Every solution of a consistent non-homogeneous system can be written as a particular solution plus a solution of the associated homogeneous system.

Exercise 2.2.33. Solve the following exercises.

- Write down the 3×3 elementary matrices corresponding to the following elementary row operations:
 - Interchanging the first and third rows ($R_1 \leftrightarrow R_3$).
 - Multiplying the second row by -4 ($R_2 \rightarrow -4R_2$).
 - Adding 3 times the second row to the first row ($R_1 \rightarrow R_1 + 3R_2$).
- Prove that every elementary matrix is non-singular and that its inverse is also an elementary matrix of the same type. Write down the inverses of the elementary matrices obtained in Problem 1.

3. Reduce the following matrix A to Row Echelon Form (REF) using elementary row operations and determine its rank:

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -1 & 7 \\ -1 & -2 & 3 & -1 \end{bmatrix}.$$

4. Reduce the following matrix A to its Reduced Row Echelon Form (RREF):

$$A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 0 & 4 & 8 \end{bmatrix}.$$

5. Find non-singular matrices P and Q such that PAQ is in the normal canonical form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ for:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix}.$$

State the rank r of matrix A .

6. Find the normal form PAQ for the matrix:

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}.$$

7. Using elementary row operations on the augmented matrix $[A \mid I_3]$, compute A^{-1} for:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}.$$

8. Prove that a square matrix $A \in M_n(\mathbb{K})$ is non-singular if and only if A is row-equivalent to the identity matrix I_n .

9. Express the non-singular matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$ as a product of elementary matrices.

10. Express $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ as a product of elementary matrices.

11. Find the complete solution set of the homogeneous system $AX = 0$. Identify the pivot variables, the free variables, and the arbitrary parameters:

$$\begin{aligned} x_1 + 2x_2 - 3x_3 + x_4 &= 0, \\ 2x_1 + 5x_2 - 8x_3 + 6x_4 &= 0, \\ 3x_1 + 4x_2 - 5x_3 - 4x_4 &= 0. \end{aligned}$$

12. Determine the values of k for which the following homogeneous system has non-trivial solutions, and find the corresponding solutions:

$$(1 - k)x_1 + 2x_2 + 3x_3 = 0,$$

$$2x_1 + (4 - k)x_2 + 6x_3 = 0,$$

$$3x_1 + 6x_2 + (9 - k)x_3 = 0.$$

13. Show that a homogeneous system $AX = 0$ with m equations and n variables has a non-trivial solution if $n > m$. If $\text{rank}(A) = r$, how many free variables occur in the general solution?
14. Test the consistency of the following non-homogeneous system using rank criteria ($\text{rank}(A) = \text{rank}([A \mid B])$), and solve if consistent:

$$x_1 + x_2 + x_3 = 6,$$

$$x_1 + 2x_2 + 3x_3 = 10,$$

$$x_1 + 2x_2 + 4x_3 = 11.$$

15. Investigate for what values of λ and μ the system:

$$x + y + z = 6,$$

$$x + 2y + 3z = 10,$$

$$x + 2y + \lambda z = \mu,$$

has: (i) no solution, (ii) a unique solution, (iii) infinitely many solutions.

16. Solve the following system completely, or show that it is inconsistent:

$$x_1 - 2x_2 + x_3 + x_4 = 2,$$

$$3x_1 + 0x_2 + 2x_3 + 2x_4 = 7,$$

$$2x_1 - 4x_2 + 2x_3 + 2x_4 = 4,$$

$$4x_1 - 2x_2 + 3x_3 + 3x_4 = 8.$$

17. Let X_p be a particular solution of $AX = B$, and let $V_0 = \{X \in \mathbb{K}^n \mid AX = 0\}$ be the solution space of the corresponding homogeneous system. Prove that the complete solution set of $AX = B$ is given by $S = \{X_p + X_h \mid X_h \in V_0\}$.
18. Solve the following system of linear equations using Gauss-Jordan elimination:

$$2x_1 + x_2 + 5x_3 = 4,$$

$$x_1 + 3x_2 + 2x_3 = 7,$$

$$3x_1 + 4x_2 + 7x_3 = 11.$$

19. Given $A \in M_n(\mathbb{K})$, suppose there exist non-singular matrices P and Q such that $PAQ = I_n$. Show that A is invertible and express A^{-1} explicitly in terms of P and Q .

20. Determine whether $B = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$ lies in the column space of $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 0 \end{bmatrix}$ by testing the consistency of $AX = B$.

Chapter 3

LINEAR COMBINATIONS, RANK AND TRACE

This unit develops the basic ideas needed to study vectors and matrices in $\mathbb{R}^n$ and $\mathbb{C}^n$. We begin with linear combinations and span and then introduce linear dependence and independence. These ideas lead naturally to basis and rank and are later used in the study of systems of linear equations and the trace of a matrix.

3.1 Linear Combinations and Span

Let $\mathbb{K}$ denote either $\mathbb{R}$ or $\mathbb{C}$. The set $\mathbb{K}^n$ consists of all column vectors with n entries from $\mathbb{K}$, that is,

$$\mathbb{K}^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{K} \right\}.$$

Thus $\mathbb{R}^n$ consists of real vectors and $\mathbb{C}^n$ consists of complex vectors.

3.1.1 Linear Combinations

Definition 3.1.1 (Linear Combination). Let $v_1, v_2, \dots, v_k \in \mathbb{K}^n$. A vector $v \in \mathbb{K}^n$ is called a **linear combination** of $v_1, v_2, \dots, v_k$ if there exist scalars $c_1, c_2, \dots, c_k \in \mathbb{K}$ such that $v = c_1 v_1 + c_2 v_2 + \dots + c_k v_k$. The scalars $c_1, c_2, \dots, c_k$ are called the **coefficients** of the linear combination.

To decide whether a vector v is a linear combination of $v_1, \dots, v_k$, place the given vectors as the columns of the matrix $A = [v_1 \ v_2 \ \dots \ v_k]$ and let $C = [c_1 \ c_2 \ \dots \ c_k]^T$. Then $c_1 v_1 + \dots + c_k v_k = v$ is equivalent to $AC = v$. Hence v is a linear combination of $v_1, \dots, v_k$ exactly when this system is consistent.

Example 3.1.2. Determine whether $v = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ is a linear combination of $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

Solution. We seek scalars c_1, c_2 such that

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}.$$

Equating corresponding components gives $c_1 = 3$, $c_2 = 1$, and $c_1 + c_2 = 2$. The first two equations give $c_1 + c_2 = 4$, which contradicts the third equation. Therefore no such scalars exist, and v is not a linear combination of v_1 and v_2 .

The same conclusion follows from the augmented matrix

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{array} \right],$$

whose last row represents the impossible equation $0 = -2$. ■

Example 3.1.3. Express $v = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}$ as a linear combination of $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

Solution. Let $v = c_1v_1 + c_2v_2 + c_3v_3$. Then

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}.$$

Equating components gives $c_1 + c_3 = 3$, $c_1 + c_2 = 5$, and $c_2 + c_3 = 4$. From the first two equations, $c_3 = 3 - c_1$ and $c_2 = 5 - c_1$. Substituting these into the third equation gives $(5 - c_1) + (3 - c_1) = 4$, so $2c_1 = 4$ and $c_1 = 2$. Therefore $c_2 = 3$ and $c_3 = 1$.

Hence $v = 2v_1 + 3v_2 + v_3$. ■

Example 3.1.4. Find all values of a for which $v = \begin{bmatrix} 1 \\ a \\ 3 \end{bmatrix}$ is a linear combination of $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

Solution. We require

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ a \\ 3 \end{bmatrix}.$$

The component equations are $c_1 = 1$, $c_1 + c_2 = a$, and $c_1 + 2c_2 = 3$. From $c_1 = 1$, the third equation gives $1 + 2c_2 = 3$, so $c_2 = 1$. Hence $a = c_1 + c_2 = 2$.

Therefore v is a linear combination of v_1 and v_2 exactly when $a = 2$. ■

3.1.2 Linear Span

Definition 3.1.5 (Linear Span). Let $S = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{K}^n$. The **linear span** of S , denoted by $\text{span}(S)$ or $\text{span}\{v_1, \dots, v_k\}$, is the set of all linear combinations of the vectors in S : $\text{span}(S) = \{c_1v_1 + \dots + c_kv_k : c_1, \dots, c_k \in \mathbb{K}\}$.

Thus $\text{span}(S)$ consists of every vector that can be obtained from the vectors in S by scalar multiplication and addition.

Definition 3.1.6 (Spanning Set). A set $S = \{v_1, v_2, \dots, v_k\}$ is said to **span** $\mathbb{K}^n$ if every vector in $\mathbb{K}^n$ can be written as a linear combination of the vectors in S ; equivalently, $\text{span}(S) = \mathbb{K}^n$.

Example 3.1.7. Show that the standard vectors $e_1, e_2, \dots, e_n$ span $\mathbb{R}^n$.

Solution. Let $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$. Since $e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$, $\dots$, $e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$, we have $x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n$. Thus every vector in $\mathbb{R}^n$ is a linear combination of $e_1, \dots, e_n$, so $\mathbb{R}^n = \text{span}\{e_1, \dots, e_n\}$. ■

Example 3.1.8. Show that $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ span $\mathbb{R}^3$.

Solution. Let $b = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3$ be arbitrary. We seek $c_1, c_2, c_3 \in \mathbb{R}$ such that $c_1 v_1 + c_2 v_2 + c_3 v_3 = b$.

Equating components gives $c_1 + c_3 = x$, $c_1 + c_2 = y$, and $c_2 + c_3 = z$.

From the first equation, $c_3 = x - c_1$, and from the second, $c_2 = y - c_1$. Substituting into the third equation gives $(y - c_1) + (x - c_1) = z$, so $2c_1 = x + y - z$. Hence

$$c_1 = \frac{x + y - z}{2}, \quad c_2 = \frac{-x + y + z}{2}, \quad c_3 = \frac{x - y + z}{2}.$$

These values exist for every $x, y, z \in \mathbb{R}$. Therefore every vector in $\mathbb{R}^3$ is a linear combination of v_1, v_2, v_3 , and hence $\mathbb{R}^3 = \text{span}\{v_1, v_2, v_3\}$. ■

Example 3.1.9. Determine whether $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ span $\mathbb{R}^3$.

Solution. A general linear combination is

$$c_1 v_1 + c_2 v_2 = \begin{bmatrix} c_1 \\ c_1 + c_2 \\ c_2 \end{bmatrix}.$$

Such a vector always satisfies $y = x + z$. Hence only vectors of the form $\begin{bmatrix} x \\ x + z \\ z \end{bmatrix}$ can be

obtained. For example, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ does not satisfy $0 = 1 + 0$, so it cannot be written as a linear combination of v_1 and v_2 .

Therefore v_1 and v_2 do not span $\mathbb{R}^3$. ■

Example 3.1.10. Show that $\mathbb{C}^2 = \text{span}\{v_1, v_2\}$ over $\mathbb{C}$, where $v_1 = \begin{bmatrix} 1 + i \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} i \\ 1 \end{bmatrix}$.

Solution. Let $w = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \in \mathbb{C}^2$. We seek $c_1, c_2 \in \mathbb{C}$ such that

$$c_1 \begin{bmatrix} 1 + i \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}.$$

Equating components gives $(1 + i)c_1 + ic_2 = z_1$ and $c_2 = z_2$. Hence $(1 + i)c_1 = z_1 - iz_2$, so $c_1 = \frac{z_1 - iz_2}{1 + i}$. Since $1 + i \neq 0$, these coefficients exist for every $z_1, z_2 \in \mathbb{C}$. Therefore $\mathbb{C}^2 = \text{span}\{v_1, v_2\}$. ■

3.1.3 Linear Dependence and Independence

Definition 3.1.11 (Linear Dependence and Independence). Let $S = \{v_1, v_2, \dots, v_k\} \subset \mathbb{K}^n$.

- (i) S is **linearly dependent** if there exist scalars $c_1, \dots, c_k$, not all zero, such that $c_1v_1 + \dots + c_kv_k = 0$.
- (ii) S is **linearly independent** if the equation $c_1v_1 + \dots + c_kv_k = 0$ has only the trivial solution $c_1 = \dots = c_k = 0$.

The phrase “not all zero” is essential. The choice $c_1 = \dots = c_k = 0$ always gives the zero vector. Linear dependence means that some other choice of coefficients also gives the zero vector.

Theorem 3.1.12 (Single-Vector Sets). For a set $S = \{v\}$, the set is linearly dependent when $v = 0$ and linearly independent when $v \neq 0$.

Proof. If $v = 0$, then $1 \cdot v = 0$ is a non-trivial relation. If $v \neq 0$ and $cv = 0$, then necessarily $c = 0$. Hence the result follows. ■

Theorem 3.1.13 (Two-Vector Sets). A set $\{v_1, v_2\}$ is linearly dependent if and only if one vector is a scalar multiple of the other.

Proof. Suppose $c_1v_1 + c_2v_2 = 0$ for some c_1, c_2 , not both zero. If $c_1 \neq 0$, then $v_1 = -\frac{c_2}{c_1}v_2$. If $c_1 = 0$, then $c_2 \neq 0$ and $v_2 = 0$, which is a scalar multiple of v_1 . Conversely, if $v_1 = cv_2$, then $v_1 - cv_2 = 0$ is a non-trivial relation. ■

Remark 3.1.14. For two non-zero vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$, linear dependence means that the vectors are parallel; linear independence means that they point in different directions.

Theorem 3.1.15 (Sets Containing the Zero Vector). Any set of vectors that contains the zero vector is linearly dependent.

Proof. If $v_1 = 0$, then $1 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_k = 0$ is a non-trivial linear relation. ■

Theorem 3.1.16 (Subset and Superset Properties). Let S be a finite set of vectors.

- (i) Every non-empty subset of a linearly independent set is linearly independent.
- (ii) Every set containing a linearly dependent subset is linearly dependent.

Proof. For (i), if a subset were linearly dependent, its non-trivial relation would also be a non-trivial relation among the vectors of S , contradicting the independence of S . For (ii), a non-trivial relation among the vectors of the dependent subset remains valid in the larger set by assigning coefficient 0 to every additional vector. ■

To test linear independence directly, solve $c_1v_1 + \dots + c_kv_k = 0$. If the only solution is $c_1 = \dots = c_k = 0$, the vectors are linearly independent; otherwise they are linearly dependent.

Example 3.1.17. Determine whether $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ are linearly independent.

Solution. Set $c_1v_1 + c_2v_2 + c_3v_3 = 0$. Then

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Equating components gives $c_1 + c_3 = 0$, $c_1 + c_2 = 0$, and $c_2 + c_3 = 0$. From the first two equations, $c_3 = -c_1$ and $c_2 = -c_1$. Substituting in the third gives $-2c_1 = 0$, so $c_1 = 0$. Hence $c_2 = c_3 = 0$. Therefore the vectors are linearly independent. ■

Example 3.1.18. Determine whether $v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$ are linearly independent.

Solution. Observe that $v_3 = v_1 + v_2$. Hence $v_1 + v_2 - v_3 = 0$ is a non-trivial linear relation. Therefore the vectors are linearly dependent. ■

Example 3.1.19. Determine whether $v_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$ and $v_2 = \begin{bmatrix} i \\ 1 \end{bmatrix}$ are linearly independent over $\mathbb{C}$.

Solution. Let $c_1 v_1 + c_2 v_2 = 0$. Then $c_1 + ic_2 = 0$ and $ic_1 + c_2 = 0$. From the first equation, $c_1 = -ic_2$. Substituting in the second gives $i(-ic_2) + c_2 = 0$, so $2c_2 = 0$. Hence $c_2 = 0$ and therefore $c_1 = 0$. Thus the vectors are linearly independent over $\mathbb{C}$. ■

Example 3.1.20. Determine whether $v_1 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -6 \\ 4 \end{bmatrix}$ are linearly independent.

Solution. Since $v_2 = -2v_1$, we have $2v_1 + v_2 = 0$. This is a non-trivial relation, so $\{v_1, v_2\}$ is linearly dependent. ■

Theorem 3.1.21 (Matrix Test for Linear Independence). *Let $v_1, v_2, \dots, v_k \in \mathbb{K}^n$ and let $A = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}$. Then $\{v_1, v_2, \dots, v_k\}$ is linearly independent if and only if the homogeneous system $AC = 0$ has only the trivial solution. Equivalently, $\text{rank}(A) = k$.*

Proof. Let $C = \begin{bmatrix} c_1 & c_2 & \cdots & c_k \end{bmatrix}^T$. By matrix multiplication, $AC = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$. Hence $AC = 0$ is exactly the equation used in the definition of linear independence. Thus the vectors are linearly independent precisely when $AC = 0$ has only the trivial solution.

After reducing A to row echelon form, $AC = 0$ has only the trivial solution exactly when there are no free variables among $c_1, \dots, c_k$. This requires a pivot in every column of A , so there are k pivots. Hence $\text{rank}(A) = k$. ■

Theorem 3.1.22 (Size Criterion for Linear Dependence). *Any set of k vectors in $\mathbb{K}^n$ with $k > n$ is linearly dependent.*

Proof. Form the matrix $A = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$ of size $n \times k$. Since there can be at most n pivots but $k > n$, at least one of the variables in $AC = 0$ is free. Hence the system has a non-trivial solution, so the vectors are linearly dependent. ■

Example 3.1.23. Use the matrix test to determine whether $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$ are linearly independent.

Solution. Place the vectors as columns:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 7 & 4 \end{bmatrix}.$$

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Now apply $R_3 \rightarrow R_3 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There are only two pivots, so $\text{rank}(A) = 2 < 3$. Hence the vectors are linearly dependent.

To find a relation, solve $AC = 0$. From the second row, $c_2 + c_3 = 0$, so $c_2 = -c_3$. The first row gives $c_1 + 2c_2 + c_3 = 0$. Substituting $c_2 = -c_3$ gives $c_1 = c_3$. Taking $c_3 = 1$ gives $c_1 = 1$ and $c_2 = -1$. Therefore $v_1 - v_2 + v_3 = 0$. ■

3.1.4 Basis and Coordinates

The ideas of spanning and linear independence come together in the concept of a basis. A basis is large enough to generate every vector in the space, but contains no redundant vector.

Definition 3.1.24 (Basis). A finite set $\mathcal{B} = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{K}^n$ is called a **basis** of $\mathbb{K}^n$ if:

- (i) $\mathcal{B}$ spans $\mathbb{K}^n$; and
- (ii) $\mathcal{B}$ is linearly independent.

Remark 3.1.25 (Standard Basis). The vectors $e_1, e_2, \dots, e_n$, where e_i has 1 in its i th position and 0 elsewhere, form the **standard basis** of $\mathbb{K}^n$. Every vector $x = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^T$ has the representation $x = x_1e_1 + \cdots + x_ne_n$.

Theorem 3.1.26 (Basis Criterion in $\mathbb{K}^n$). Let $v_1, \dots, v_n \in \mathbb{K}^n$ and form the square matrix $A = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$. The following statements are equivalent:

- (i) $\{v_1, \dots, v_n\}$ is a basis of $\mathbb{K}^n$;
- (ii) $v_1, \dots, v_n$ are linearly independent;
- (iii) $v_1, \dots, v_n$ span $\mathbb{K}^n$;
- (iv) $\text{rank}(A) = n$.

Proof. The vectors are linearly independent exactly when the homogeneous system $AC = 0$ has only the trivial solution. This is equivalent to a pivot in every column of A , and hence to $\text{rank}(A) = n$. The vectors span $\mathbb{K}^n$ exactly when $AC = b$ is consistent for every $b \in \mathbb{K}^n$. Since A is square, this is equivalent to a pivot in every row, again giving $\text{rank}(A) = n$. Therefore all four statements are equivalent. ■

Theorem 3.1.27 (Number of Vectors in a Basis). Every basis of $\mathbb{K}^n$ contains exactly n vectors. Consequently, the dimension of $\mathbb{K}^n$ is n .

Proof. Let $\{v_1, \dots, v_k\}$ be a basis of $\mathbb{K}^n$. Since the set is linearly independent, the size criterion gives $k \leq n$. Since it spans $\mathbb{K}^n$, the matrix $A = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$ must have rank n , so it must have at least n columns; hence $k \geq n$. Therefore $k = n$. ■

Theorem 3.1.28 (Unique Representation). If $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis of $\mathbb{K}^n$, then every vector $x \in \mathbb{K}^n$ can be written in exactly one way as $x = c_1v_1 + \cdots + c_nv_n$.

Proof. Because $\mathcal{B}$ spans $\mathbb{K}^n$, at least one representation exists. Suppose $x = c_1v_1 + \cdots + c_nv_n = d_1v_1 + \cdots + d_nv_n$. Subtracting gives $(c_1 - d_1)v_1 + \cdots + (c_n - d_n)v_n = 0$. Since the basis vectors are linearly independent, $c_i - d_i = 0$ for every i . Hence $c_i = d_i$ for all i , so the representation is unique. ■

Definition 3.1.29 (Coordinates Relative to an Ordered Basis). Let $\mathcal{B} = (v_1, \dots, v_n)$ be an ordered basis of $\mathbb{K}^n$. If $x = c_1v_1 + \cdots + c_nv_n$, then $[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$ is called the **coordinate vector of x relative to $\mathcal{B}$** .

Example 3.1.30. Let $\mathcal{B} = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right)$. Find the coordinates of $x = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$ relative to $\mathcal{B}$.

Solution. We seek c_1, c_2, c_3 such that $x = c_1v_1 + c_2v_2 + c_3v_3$. Thus

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}.$$

Equating components gives $c_1 + c_2 + c_3 = 4$, $c_2 + c_3 = 1$, and $c_3 = 2$. Hence $c_3 = 2$, $c_2 = -1$, and $c_1 = 3$. Therefore $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$. ■

Example 3.1.31. Determine whether $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}$ is a basis of $\mathbb{R}^3$.

Solution. The third vector is the sum of the first two, since $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Therefore the vectors are linearly dependent. A basis must be linearly independent, so this set is not a basis of $\mathbb{R}^3$. ■

3.2 Rank of a Matrix and Solution Space of Linear Systems

The rank of a matrix measures the number of linearly independent rows or columns that it contains. It can be found efficiently by row reduction and is closely related to the number of free variables in a homogeneous system.

3.2.1 Rank of a Matrix

Definition 3.2.1 (Row Space and Column Space). Let $A \in M_{m \times n}(\mathbb{K})$.

- (i) The **row space** of A , denoted by $\text{row}(A)$, is the span of the rows of A .
- (ii) The **column space** of A , denoted by $\text{col}(A)$, is the span of the columns of A .

Theorem 3.2.2 (Equality of Row Rank and Column Rank). *For every matrix A , the maximum number of linearly independent rows equals the maximum number of linearly independent columns.*

This common number is called the rank of the matrix.

Definition 3.2.3 (Rank). The **rank** of a matrix A , denoted by $\text{rank}(A)$, is the common dimension of its row space and column space. Equivalently, it is the maximum number of linearly independent rows, or the maximum number of linearly independent columns, of A .

Theorem 3.2.4 (Preservation of Rank under Elementary Row Operations). *Elementary row operations preserve the row space and hence preserve the rank of a matrix.*

Proof. Let the rows of A be $R_1, \dots, R_m$. A row interchange merely changes their order. Under $R_i \rightarrow cR_i$, where $c \neq 0$, the old row can be recovered as $R_i = c^{-1}(cR_i)$. Under $R_i \rightarrow R_i + cR_j$, the old row can be recovered as $R_i = (R_i + cR_j) - cR_j$. Thus each elementary row operation can be reversed, and the old and new rows span the same row space. Therefore the rank is unchanged. ■

Corollary 3.2.5 (Rank from Echelon Form). *The rank of a matrix equals the number of non-zero rows, or equivalently the number of pivot positions, in any row echelon form of the matrix.*

This gives the standard procedure for computing rank: reduce the matrix to REF or RREF and count its pivots.

Example 3.2.6. Find the rank of

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 0 & 5 \end{bmatrix}.$$

Solution. Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 0 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -4 \\ 0 & 0 & 3 & -4 \end{bmatrix}.$$

Now apply $R_3 \rightarrow R_3 - R_2$:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are two non-zero rows, hence two pivots. Therefore $\text{rank}(A) = 2$. ■

Example 3.2.7. Find the rank of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 7 & 11 \end{bmatrix}.$$

Solution. Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 7 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}.$$

Then apply $R_3 \rightarrow R_3 - R_2$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $\text{rank}(A) = 2$. ■

Example 3.2.8. Determine the value of k for which

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & k \end{bmatrix}$$

has rank 1, and determine its rank for all other values of k .

Solution. Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & k-1 \end{bmatrix}.$$

If $k = 1$, the last row is zero, so only one non-zero row remains and $\text{rank}(A) = 1$. If $k \neq 1$, the last row is non-zero, giving a second pivot. Hence $\text{rank}(A) = 2$. ■

3.2.2 Rank and the Homogeneous System $AX = 0$

Consider a homogeneous system $AX = 0$, where $A \in M_{m \times n}(\mathbb{K})$. If $\text{rank}(A) = r$, then the RREF of A has exactly r pivot columns. Hence r variables are pivot variables and the remaining $n - r$ variables are free.

Theorem 3.2.9 (Dimension of the Solution Space). *Let $A \in M_{m \times n}(\mathbb{K})$ have rank r . Then the solution space of $AX = 0$ has dimension $n - r$.*

Proof. Row operations do not change the solutions of a homogeneous system, so $AX = 0$ and $\text{RREF}(A)X = 0$ have the same solution set. Since $\text{rank}(A) = r$, there are r pivot variables and $n - r$ free variables.

Assign parameters $t_1, \dots, t_{n-r}$ to the free variables. The pivot variables can then be expressed in terms of these parameters, so every solution can be written in the form $X = t_1 X_1 + \dots + t_{n-r} X_{n-r}$, where the vectors $X_1, \dots, X_{n-r}$ are obtained by setting one free parameter equal to 1 and the others equal to 0. These vectors form a basis of the solution space. Therefore its dimension is $n - r$. ■

Remark 3.2.10. The identity $\text{rank}(A) + \dim N(A) = n$ is called the **rank-nullity relation**.

Corollary 3.2.11. *For a homogeneous system $AX = 0$ with n variables:*

- (i) *if $\text{rank}(A) = n$, there are no free variables and the system has only the trivial solution;*
- (ii) *if $\text{rank}(A) < n$, at least one free variable occurs and the system has non-trivial solutions.*

Example 3.2.12. For

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 3 \\ 3 & 6 & 2 & 7 \end{bmatrix},$$

find $\text{rank}(A)$ and the complete solution of $AX = 0$. Hence find a basis of the solution space.

Solution. First reduce A . Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -5 \\ 0 & 0 & 5 & -5 \end{bmatrix}.$$

Now apply $R_3 \rightarrow R_3 - R_2$ and $R_2 \rightarrow \frac{1}{5}R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Finally apply $R_1 \rightarrow R_1 + R_2$:

$$\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are two pivots, so $\text{rank}(A) = 2$. Since there are four variables, x_2 and x_4 are free.

The equations are $x_1 + 2x_2 + 3x_4 = 0$ and $x_3 - x_4 = 0$. Hence $x_1 = -2x_2 - 3x_4$ and $x_3 = x_4$. Let $x_2 = s$ and $x_4 = t$. Then

$$X = \begin{bmatrix} -2s - 3t \\ s \\ t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Thus a basis of the solution space is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\},$$

and its dimension is $4 - 2 = 2$. ■

Example 3.2.13. Determine the values of λ for which the homogeneous system

$$\begin{aligned} x + y + z &= 0, \\ x + \lambda y + z &= 0, \\ x + y + \lambda z &= 0 \end{aligned}$$

has a non-trivial solution.

Solution. The coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{bmatrix}.$$

A non-trivial solution exists exactly when $\text{rank}(A) < 3$, or equivalently when $\det A = 0$. Subtract R_1 from R_2 and R_3 :

$$\det A = \begin{vmatrix} 1 & 1 & 1 \\ 0 & \lambda - 1 & 0 \\ 0 & 0 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2.$$

Hence $\det A = 0$ precisely when $\lambda = 1$. Therefore the system has a non-trivial solution only for $\lambda = 1$.

When $\lambda = 1$, all three equations reduce to $x + y + z = 0$. Let $y = s$ and $z = t$. Then $x = -s - t$, so

$$X = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}. \quad \text{■}$$

3.2.3 Further Properties of Rank

Definition 3.2.14 (Rank in Terms of Minors). The rank of a matrix is the order of its largest square submatrix having non-zero determinant. Equivalently, $\text{rank}(A) = r$ if A has at least one non-zero minor of order r and every minor of order greater than r is zero.

Example 3.2.15. Find the rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ using minors.

Solution. Since

$$\det A = 0,$$

the rank is less than 3. Consider the 2×2 minor formed from rows 1, 3 and columns 1, 2:

$$\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 - 2 = -1 \neq 0.$$

Hence a non-zero minor of order 2 exists. Therefore $\text{rank}(A) = 2$. ■

Theorem 3.2.16 (Rank of a Product). If $A \in M_{m \times p}(\mathbb{K})$ and $B \in M_{p \times n}(\mathbb{K})$, then

$$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

Proof. Every column of AB has the form Ab_j , where b_j is a column of B . Hence every column of AB is a linear combination of the columns of A , so $\text{rank}(AB) \leq \text{rank}(A)$.

Similarly, every row of AB is a linear combination of the rows of B , so $\text{rank}(AB) \leq \text{rank}(B)$. Combining the two inequalities gives the result. ■

Remark 3.2.17. The inequality above is often called the **rank inequality for a product**. The term **Sylvester rank inequality** usually refers to the stronger lower bound $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - p$ when A has p columns and B has p rows.

Example 3.2.18. Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 3 \end{bmatrix}.$$

Verify the rank inequality for AB .

Solution. Both rows of A are linearly independent, so $\text{rank}(A) = 2$. The two columns of B are not scalar multiples of one another, so $\text{rank}(B) = 2$.

Now

$$AB = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}.$$

Since $\det(AB) = 8 - 5 = 3 \neq 0$, $\text{rank}(AB) = 2$. Therefore

$$\text{rank}(AB) = 2 \leq \min\{2, 2\} = 2.$$

In this example equality holds. ■

Example 3.2.19. Give an example where $\text{rank}(AB)$ is strictly smaller than both $\text{rank}(A)$ and $\text{rank}(B)$.

Solution. Take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Both A and B have rank 1, but

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

so $\text{rank}(AB) = 0 < 1$. ■

Theorem 3.2.20 (Invariance of Rank under Non-Singular Multiplication). *Let $A \in M_{m \times n}(\mathbb{K})$. If $P \in M_m(\mathbb{K})$ and $Q \in M_n(\mathbb{K})$ are non-singular, then*

$$\text{rank}(PA) = \text{rank}(A) = \text{rank}(AQ).$$

Proof. By the rank inequality for a product, $\text{rank}(AQ) \leq \text{rank}(A)$. Since Q is non-singular, $A = (AQ)Q^{-1}$, and therefore $\text{rank}(A) \leq \text{rank}(AQ)$. Hence $\text{rank}(AQ) = \text{rank}(A)$.

Similarly, $\text{rank}(PA) \leq \text{rank}(A)$, while $A = P^{-1}(PA)$ gives $\text{rank}(A) \leq \text{rank}(PA)$. Thus $\text{rank}(PA) = \text{rank}(A)$. ■

Corollary 3.2.21. *If P and Q are non-singular, then $\text{rank}(PAQ) = \text{rank}(A)$.*

3.3 Trace of a Matrix and Its Properties

The trace of a square matrix is the sum of its diagonal entries. Although simple to compute, it has several important algebraic properties, especially in products of matrices and similarity transformations.

Definition 3.3.1 (Trace of a Matrix). Let $A = (a_{ij}) \in M_n(\mathbb{K})$. The **trace** of A , denoted by $\text{Tr}(A)$ or $\text{tr}(A)$, is

$$\text{Tr}(A) = a_{11} + a_{22} + \cdots + a_{nn} = \sum_{i=1}^n a_{ii}.$$

The trace is defined only for square matrices.

Example 3.3.2. Find the trace of $A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & 5 & 0 \\ 1 & 7 & -2 \end{bmatrix}$.

Solution. The diagonal entries are 3, 5, -2 . Hence $\text{Tr}(A) = 3 + 5 - 2 = 6$. ■

Theorem 3.3.3 (Basic Properties of Trace). *Let $A, B \in M_n(\mathbb{K})$ and $c \in \mathbb{K}$. Then:*

- (i) $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$;
- (ii) $\text{Tr}(cA) = c \text{Tr}(A)$;
- (iii) $\text{Tr}(A^T) = \text{Tr}(A)$;
- (iv) if $A \in M_n(\mathbb{C})$, then $\text{Tr}(A^*) = \overline{\text{Tr}(A)}$;
- (v) $\text{Tr}(I_n) = n$.

Proof. Let $A = (a_{ij})$ and $B = (b_{ij})$.

(i) The diagonal entries of $A + B$ are $a_{ii} + b_{ii}$, so $\text{Tr}(A + B) = \sum_{i=1}^n (a_{ii} + b_{ii}) = \text{Tr}(A) + \text{Tr}(B)$.

(ii) The diagonal entries of cA are ca_{ii} , so $\text{Tr}(cA) = \sum_{i=1}^n ca_{ii} = c \text{Tr}(A)$.

- (iii) Transposition does not change the diagonal entries, hence $\text{Tr}(A^T) = \text{Tr}(A)$.
- (iv) The diagonal entries of A^* are $\overline{a_{ii}}$, so $\text{Tr}(A^*) = \sum_{i=1}^n \overline{a_{ii}} = \overline{\text{Tr}(A)}$.
- (v) Since every diagonal entry of I_n is 1, $\text{Tr}(I_n) = n$.

■

Example 3.3.4. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 0 & 5 \end{bmatrix}$. Verify the linearity of trace for $2A - B$.

Solution. We have $\text{Tr}(A) = 1 + 4 = 5$ and $\text{Tr}(B) = 2 + 5 = 7$. Therefore

$$2\text{Tr}(A) - \text{Tr}(B) = 2(5) - 7 = 3.$$

Also,

$$2A - B = \begin{bmatrix} 0 & 5 \\ 6 & 3 \end{bmatrix},$$

so $\text{Tr}(2A - B) = 0 + 3 = 3$. Hence $\text{Tr}(2A - B) = 2\text{Tr}(A) - \text{Tr}(B)$.

■

3.3.1 The Identity $\text{Tr}(AB) = \text{Tr}(BA)$

A particularly useful property of trace is that two factors may be interchanged inside the trace, even though matrix multiplication itself is generally not commutative.

Theorem 3.3.5 (Trace of a Product). *Let $A \in M_{m \times n}(\mathbb{K})$ and $B \in M_{n \times m}(\mathbb{K})$. Then*

$$\text{Tr}(AB) = \text{Tr}(BA).$$

Proof. Let $A = (a_{ik})$ and $B = (b_{ki})$. The i th diagonal entry of AB is $(AB)_{ii} = \sum_{k=1}^n a_{ik}b_{ki}$. Hence

$$\text{Tr}(AB) = \sum_{i=1}^m \sum_{k=1}^n a_{ik}b_{ki}.$$

Similarly, $(BA)_{kk} = \sum_{i=1}^m b_{ki}a_{ik}$, so

$$\text{Tr}(BA) = \sum_{k=1}^n \sum_{i=1}^m b_{ki}a_{ik}.$$

Since these are finite sums of the same scalar products, $\text{Tr}(AB) = \text{Tr}(BA)$.

■

Remark 3.3.6. The products AB and BA need not have the same size. The theorem applies whenever both products are defined and square.

Example 3.3.7. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$. Verify that $\text{Tr}(AB) = \text{Tr}(BA)$.

Solution. First,

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 16 \end{bmatrix}.$$

Thus $\text{Tr}(AB) = 2 + 16 = 18$.

Next,

$$BA = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 4 \\ -3 & 1 & -4 \\ 10 & -1 & 12 \end{bmatrix}.$$

Hence $\text{Tr}(BA) = 5 + 1 + 12 = 18$. Therefore $\text{Tr}(AB) = \text{Tr}(BA)$.

■

Corollary 3.3.8 (Cyclic Property). *For conformable matrices A, B, C for which the products are square,*

$$\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB).$$

Proof. Apply $\text{Tr}(XY) = \text{Tr}(YX)$ with $X = A$ and $Y = BC$ to obtain $\text{Tr}(ABC) = \text{Tr}(BCA)$. Applying it with $X = AB$ and $Y = C$ gives $\text{Tr}(ABC) = \text{Tr}(CAB)$. ■

Remark 3.3.9. Only cyclic rearrangements are guaranteed. In general, $\text{Tr}(ABC)$ need not equal $\text{Tr}(BAC)$.

Example 3.3.10. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$. Verify the cyclic property.

Solution. We compute

$$ABC = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix},$$

so $\text{Tr}(ABC) = 9$.

Also,

$$BCA = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & 7 \end{bmatrix},$$

so $\text{Tr}(BCA) = 9$.

Finally,

$$CAB = \begin{bmatrix} 1 & 1 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 6 & 6 \end{bmatrix},$$

so $\text{Tr}(CAB) = 9$. Hence all three traces are equal. ■

Theorem 3.3.11 (Similarity Invariance). *If $A, B \in M_n(\mathbb{K})$ and $B = P^{-1}AP$ for some non-singular matrix P , then $\text{Tr}(B) = \text{Tr}(A)$.*

Proof. Using the product identity,

$$\text{Tr}(B) = \text{Tr}(P^{-1}AP) = \text{Tr}(APP^{-1}) = \text{Tr}(A).$$

Thus similar matrices have the same trace. ■

Example 3.3.12. Let $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$. Find $B = P^{-1}AP$ and verify that $\text{Tr}(B) = \text{Tr}(A)$.

Solution. Since $\det P = 1$, we have $P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$. First compute

$$AP = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 6 & 10 \end{bmatrix}.$$

Therefore

$$B = P^{-1}(AP) = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 2 & 5 \end{bmatrix}.$$

Now $\text{Tr}(A) = 3 + 4 = 7$ and $\text{Tr}(B) = 2 + 5 = 7$. Hence $\text{Tr}(B) = \text{Tr}(A)$. ■

Example 3.3.13. Show that there are no matrices $A, B \in M_n(\mathbb{R})$ or $M_n(\mathbb{C})$ satisfying $AB - BA = I_n$.

Solution. Assume $AB - BA = I_n$. Taking trace on both sides gives

$$\text{Tr}(AB) - \text{Tr}(BA) = \text{Tr}(I_n).$$

Since $\text{Tr}(AB) = \text{Tr}(BA)$, the left-hand side is 0, while $\text{Tr}(I_n) = n$. Thus $0 = n$, which is impossible for $n \geq 1$. Hence no such matrices exist. ■

Example 3.3.14. If $A, B \in M_n(\mathbb{K})$, show that $\text{Tr}(AB - BA) = 0$.

Solution. By linearity and the product identity,

$$\text{Tr}(AB - BA) = \text{Tr}(AB) - \text{Tr}(BA) = 0.$$

■

Exercise 3.3.15. Solve the following exercises based on this unit.

1. Determine whether $v = \begin{bmatrix} 3 \\ 8 \\ 11 \end{bmatrix}$ can be expressed as a linear combination of $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$ in $\mathbb{R}^3$. If possible, find the coefficients.
2. Show that $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ spans $\mathbb{R}^3$.
3. Show that $\mathcal{B} = \left\{ \begin{bmatrix} 1+i \\ 0 \end{bmatrix}, \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathbb{C}^2$ over $\mathbb{C}$.
4. Find the coordinates of $x = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$ relative to the ordered basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ of $\mathbb{R}^3$.
5. Determine whether $v = \begin{bmatrix} 1 \\ a \\ 3 \end{bmatrix}$ is a linear combination of $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$. Find the value of a for which this is possible.
6. Using the definition directly, determine whether $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \right\} \subset \mathbb{R}^3$ is linearly independent.
7. Using the matrix rank test, determine whether $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 3 \end{bmatrix} \right\} \subset \mathbb{R}^4$ is linearly independent.
8. Prove that any set of k vectors in $\mathbb{K}^n$ with $k > n$ is linearly dependent.
9. Determine all real values of k for which $\left\{ \begin{bmatrix} 1 \\ 1 \\ k \end{bmatrix}, \begin{bmatrix} 1 \\ k \\ 1 \end{bmatrix}, \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} \right\}$ is linearly independent in $\mathbb{R}^3$.

10. Determine whether the vectors $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$ are linearly independent. If they are dependent, find a non-trivial linear relation among them.
11. Reduce $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 0 & 5 \end{bmatrix}$ to Row Echelon Form and hence determine $\text{rank}(A)$.
12. Compute the rank of $A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 2 & 3 & 7 & 11 \\ 1 & 3 & 8 & 10 \end{bmatrix}$ by row reduction.
13. Prove that elementary row operations preserve the row space and therefore preserve the rank of a matrix.
14. Prove that for every $A \in M_{m \times n}(\mathbb{K})$, $\text{rank}(A) \leq \min\{m, n\}$.
15. Determine the value of k for which $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & k \end{bmatrix}$ has rank 1. Determine its rank for all other values of k .
16. Find the rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ using minors.
17. For $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 3 \\ 3 & 6 & 2 & 7 \end{bmatrix}$:
- find $\text{rank}(A)$;
 - find the number of free variables in $AX = 0$;
 - find the general solution of $AX = 0$;
 - find a basis for the solution space.
18. State the Rank–Nullity Theorem. If A is a 4×6 matrix with $\text{rank}(A) = 3$, determine the dimension of the solution space of $AX = 0$.
19. Find the general solution and a basis for the solution space of $AX = 0$, where $A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 0 & 4 & 8 \end{bmatrix}$.
20. Prove that if $A \in M_{m \times n}(\mathbb{K})$ has $\text{rank}(A) = n$, then the homogeneous system $AX = 0$ has only the trivial solution.
21. Determine all values of λ for which the homogeneous system
- $$x + y + z = 0, \quad x + \lambda y + z = 0, \quad x + y + \lambda z = 0$$
- has a non-trivial solution. Find the general solution for the exceptional value(s) of λ .
22. Let $A \in M_{m \times p}(\mathbb{K})$ and $B \in M_{p \times n}(\mathbb{K})$. Prove that $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.
23. If P and Q are non-singular matrices of appropriate sizes, prove that $\text{rank}(PAQ) = \text{rank}(A)$.

24. Compute the trace of $A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & 5 & -2 \\ 1 & 0 & 6 \end{bmatrix}$ and verify that $\text{Tr}(A^T) = \text{Tr}(A)$.
25. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ -1 & 5 \end{bmatrix}$. Compute $\text{Tr}(3A - 2B)$ directly and verify that it agrees with $3 \text{Tr}(A) - 2 \text{Tr}(B)$.
26. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$, compute AB and BA and verify that $\text{Tr}(AB) = \text{Tr}(BA)$.
27. Prove that for conformable matrices A, B, C , $\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$.
28. Show by an example that an arbitrary interchange of factors is not valid; that is, $\text{Tr}(ABC)$ need not equal $\text{Tr}(BAC)$.
29. Prove that $\text{Tr}(AB - BA) = 0$ for all $A, B \in M_n(\mathbb{K})$.
30. Prove that there do not exist matrices $A, B \in M_n(\mathbb{R})$ or $M_n(\mathbb{C})$ satisfying $AB - BA = I_n$.
31. Prove the similarity invariance of trace: if $B = P^{-1}AP$, where P is non-singular, then $\text{Tr}(B) = \text{Tr}(A)$.
32. Let $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$. Find $B = P^{-1}AP$ and verify directly that $\text{Tr}(B) = \text{Tr}(A)$.

Chapter 4

MATRIX POLYNOMIALS AND INNER PRODUCT SPACES

This unit develops matrix polynomials, algebraic matrix functions, and the spectral properties of square matrices. It introduces characteristic and minimal polynomials, establishing eigenvalues and eigenvectors as intrinsic invariants of linear transformations. The unit examines the Cayley–Hamilton theorem alongside its practical computational applications in matrix inversion and power calculations. Finally, it investigates inner product structures on $\mathbb{R}^n$ and $\mathbb{C}^n$, establishing norms, orthogonality, orthonormal sets, and geometric interpretations of length and angles.

4.1 Matrix Polynomials and Algebraic Functions of Matrices

In scalar algebra, polynomials form the fundamental class of functions. By substituting a square matrix $A \in M_n(\mathbb{K})$ in place of a scalar variable x , we extend polynomial algebra to matrix spaces. This leads to matrix polynomials, annihilating polynomials, and formal algebraic matrix functions.

Definition 4.1.1 (Matrix Polynomial). Let $p(x) = a_k x^k + a_{k-1} x^{k-1} + \dots + a_1 x + a_0$ be a polynomial of degree k over a field $\mathbb{K}$ (where $a_k \neq 0$). For a square matrix $A \in M_n(\mathbb{K})$, the **matrix polynomial** $p(A)$ is the $n \times n$ matrix defined by: $p(A) = a_k A^k + a_{k-1} A^{k-1} + \dots + a_1 A + a_0 I_n$, where $A^0 = I_n$ is the $n \times n$ identity matrix, and A^j denotes the j -th matrix power ($A^j = \underbrace{A \cdot A \cdots A}_{j \text{ times}}$).

In scalar algebra, a_0 is a scalar. In matrix algebra, addition is defined only between matrices of identical dimensions. Therefore, the scalar constant a_0 in $p(x)$ **must** be multiplied by the identity matrix I_n when evaluating $p(A)$.

Theorem 4.1.2 (Commutativity and Homomorphism of Matrix Polynomials). Let $p(x), q(x) \in \mathbb{K}[x]$ be polynomials, $A \in M_n(\mathbb{K})$, and $c \in \mathbb{K}$.

- (i) **Additive Homomorphism:** $(p + q)(A) = p(A) + q(A)$.
- (ii) **Scalar Homomorphism:** $(c \cdot p)(A) = c \cdot p(A)$.
- (iii) **Multiplicative Homomorphism:** $(p \cdot q)(A) = p(A)q(A)$.
- (iv) **Commutativity:** Matrix polynomials in the same matrix A always commute under multiplication: $p(A)q(A) = q(A)p(A)$.

Proof. Let $p(x) = \sum_{i=0}^k a_i x^i$ and $q(x) = \sum_{j=0}^m b_j x^j$ be polynomials in $\mathbb{K}[x]$. Without loss of generality, by padding with zero coefficients if necessary, we may assume both polynomials are expressed up to degree k .

We prove the four statements in turn.

(i) *Additive homomorphism.*

The sum polynomial is $(p + q)(x) = \sum_{i=0}^k (a_i + b_i) x^i$. Evaluating at A :

$$\begin{aligned} (p + q)(A) &= \sum_{i=0}^k (a_i + b_i) A^i \\ &= \sum_{i=0}^k (a_i A^i + b_i A^i) \\ &= \sum_{i=0}^k a_i A^i + \sum_{i=0}^k b_i A^i \\ &= p(A) + q(A). \end{aligned}$$

(ii) *Scalar homomorphism.*

The scaled polynomial is $(c \cdot p)(x) = \sum_{i=0}^k (ca_i) x^i$. Evaluating at A :

$$\begin{aligned} (c \cdot p)(A) &= \sum_{i=0}^k (ca_i) A^i \\ &= c \sum_{i=0}^k a_i A^i \\ &= c \cdot p(A). \end{aligned}$$

(iii) *Multiplicative homomorphism.*

By definition of polynomial multiplication, the product polynomial is $(p \cdot q)(x) = \sum_{l=0}^{k+m} c_l x^l$, where $c_l = \sum_{i+j=l} a_i b_j$. Evaluating at A and using $A^{i+j} = A^i A^j$:

$$\begin{aligned} (p \cdot q)(A) &= \sum_{l=0}^{k+m} \left(\sum_{i+j=l} a_i b_j \right) A^{i+j} \\ &= \sum_{i=0}^k \sum_{j=0}^m a_i b_j A^i A^j \\ &= \left(\sum_{i=0}^k a_i A^i \right) \left(\sum_{j=0}^m b_j A^j \right) \\ &= p(A)q(A). \end{aligned}$$

(iv) *Commutativity.*

In scalar polynomial arithmetic, polynomial multiplication is commutative, so $(p \cdot q)(x) = (q \cdot p)(x)$ in $\mathbb{K}[x]$.

Applying the multiplicative homomorphism from Part (iii) to both sides:

$$\begin{aligned} p(A)q(A) &= (p \cdot q)(A) \\ &= (q \cdot p)(A) \quad (\text{since } p(x)q(x) = q(x)p(x) \text{ in } \mathbb{K}[x]) \\ &= q(A)p(A). \end{aligned}$$

Thus, $p(A)q(A) = q(A)p(A)$, proving that matrix polynomials in the same matrix A always commute. ■

Theorem 4.1.3 (Spectral Mapping Theorem for Matrix Polynomials). *Let $A \in M_n(\mathbb{K})$. If $\lambda \in \mathbb{K}$ is an eigenvalue of A with corresponding eigenvector $x \in \mathbb{K}^n \setminus \{0\}$, then $p(\lambda)$ is an eigenvalue of the matrix polynomial $p(A)$ with the same eigenvector x : $p(A)x = p(\lambda)x$.*

Proof. Since $Ax = \lambda x$, applying A repeatedly yields $A^i x = \lambda^i x$ for all integers $i \geq 0$.

Let $p(x) = a_k x^k + \dots + a_1 x + a_0$. Substituting A and applying $p(A)$ to x :

$$\begin{aligned} p(A)x &= (a_k A^k + \dots + a_1 A + a_0 I_n)x \\ &= a_k A^k x + \dots + a_1 A x + a_0 x \\ &= a_k \lambda^k x + \dots + a_1 \lambda x + a_0 x \\ &= (a_k \lambda^k + \dots + a_1 \lambda + a_0)x = p(\lambda)x. \end{aligned}$$

Thus, x is an eigenvector of $p(A)$ with corresponding eigenvalue $p(\lambda)$. ■

4.1.1 Annihilating Polynomials and Special Matrix Classes

Definition 4.1.4 (Annihilating Polynomial). A polynomial $p(x) \in \mathbb{K}[x]$ is called an **annihilating polynomial** (or **zeroing polynomial**) of a square matrix $A \in M_n(\mathbb{K})$ if: $p(A) = 0_{n \times n}$.

Several important classes of matrices are defined by simple polynomial algebraic relations:

1. **Idempotent Matrices (Projections):** A matrix A satisfying $A^2 = A$. Its annihilating polynomial is $p(x) = x^2 - x = x(x - 1)$.
2. **Involutory Matrices:** A matrix A satisfying $A^2 = I_n$. Its annihilating polynomial is $p(x) = x^2 - 1 = (x - 1)(x + 1)$.
3. **Nilpotent Matrices:** A matrix A for which there exists an integer $k \geq 1$ such that $A^k = 0$. Its annihilating polynomial is $p(x) = x^k$.
4. **Nilpotent Index:** The smallest positive integer k such that $A^k = 0$ is called the **index of nilpotency**.

4.1.2 Algebraic Matrix Functions and Power Series

Beyond finite polynomials, algebraic functions of matrices can be defined via infinite power series and rational expressions, provided convergence criteria or termination conditions are met.

Theorem 4.1.5 (Inverse of $(I_n - A)$ for a Nilpotent Matrix). *Let $A \in M_n(\mathbb{K})$ be a nilpotent matrix with index k (such that $A^k = 0$). Then the matrix $(I_n - A)$ is invertible, and its inverse is given by the finite polynomial sum:*

$$(I_n - A)^{-1} = I_n + A + A^2 + \dots + A^{k-1}.$$

Proof. Let $B = I_n + A + A^2 + \dots + A^{k-1}$. Multiplying $(I_n - A)$ by B expands as:

$$\begin{aligned} (I_n - A)B &= I_n(I_n + A + A^2 + \dots + A^{k-1}) - A(I_n + A + A^2 + \dots + A^{k-1}) \\ &= (I_n + A + A^2 + \dots + A^{k-1}) - (A + A^2 + A^3 + \dots + A^k). \end{aligned}$$

Notice that all intermediate terms cancel in pairs, leaving only: $(I_n - A)B = I_n - A^k$. Since A is nilpotent with $A^k = 0$, this simplifies to $(I_n - A)B = I_n$.

Similarly, $B(I_n - A) = I_n$. Therefore, $(I_n - A)$ is invertible and $(I_n - A)^{-1} = B$. ■

Definition 4.1.6 (Matrix Exponential Function). For any square complex matrix $A \in M_n(\mathbb{C})$, the **matrix exponential** e^A (or $\exp(A)$) is defined by the convergent matrix power series:

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k = I_n + A + \frac{1}{2!} A^2 + \frac{1}{3!} A^3 + \dots$$

If A is nilpotent with $A^k = 0$, the series terminates after k terms, yielding a finite matrix polynomial.

Example 4.1.7. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \in M_2(\mathbb{R})$ and $p(x) = x^2 - 4x + 3$. Evaluate $p(A)$.

Solution. We compute the matrix power A^2 :

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}.$$

Now substitute A into $p(A) = A^2 - 4A + 3I_2$:

$$\begin{aligned} p(A) &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 0 & 12 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1-4+3 & 8-8+0 \\ 0-0+0 & 9-12+3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Thus, $p(A) = 0_{2 \times 2}$, showing that $p(x) = x^2 - 4x + 3$ is an annihilating polynomial for A . ■

Example 4.1.8. Let $A = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$. Show that A is nilpotent, and compute $(I_3 - A)^{-1}$ and e^A as matrix polynomials.

Solution. We compute powers of A :

$$\begin{aligned} A^2 &= \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \\ A^3 &= A^2 A = \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Since $A^3 = 0$, A is nilpotent with index $k = 3$.

Using the algebraic series formula for $(I_3 - A)^{-1}$:

$$(I_3 - A)^{-1} = I_3 + A + A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

For the matrix exponential e^A , the infinite series terminates at $k = 3$:

$$\begin{aligned} e^A &= I_3 + A + \frac{1}{2} A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

■

Example 4.1.9. Let $A \in M_2(\mathbb{R})$ have eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$. Find the eigenvalues of the matrix polynomial $p(A) = A^3 - 2A + 4I_2$.

Solution. By the Spectral Mapping Theorem for Matrix Polynomials, if λ is an eigenvalue of A , then $p(\lambda)$ is an eigenvalue of $p(A)$, where $p(x) = x^3 - 2x + 4$.

Evaluating $p(x)$ at each eigenvalue of A :

- For $\lambda_1 = 2$: $p(\lambda_1) = p(2) = 2^3 - 2(2) + 4 = 8 - 4 + 4 = 8$.
- For $\lambda_2 = -1$: $p(\lambda_2) = p(-1) = (-1)^3 - 2(-1) + 4 = -1 + 2 + 4 = 5$.

Thus, the eigenvalues of $p(A)$ are 8 and 5. ■

4.2 Characteristic and Minimal Polynomials of a Matrix

To every square matrix $A \in M_n(\mathbb{K})$, we can associate two special polynomials: the **characteristic polynomial**, which encodes the matrix's determinantal structure and eigenvalues, and the **minimal polynomial**, which represents the simplest algebraic equation satisfied by the matrix.

Definition 4.2.1 (Characteristic Matrix and Polynomial). Let $A \in M_n(\mathbb{K})$ be an $n \times n$ matrix over a field $\mathbb{K}$, and let λ be an indeterminate scalar.

- The matrix $\lambda I_n - A$ is called the **characteristic matrix** of A .
- The determinant of the characteristic matrix, denoted by $p_A(\lambda)$ or $\chi_A(\lambda)$, is a polynomial in λ of degree n called the **characteristic polynomial** of A :

$$p_A(\lambda) = \det(\lambda I_n - A).$$

Theorem 4.2.2 (Structure of the Characteristic Polynomial). For any $n \times n$ matrix $A \in M_n(\mathbb{K})$, the characteristic polynomial $p_A(\lambda)$ has the general expansion:

$$p_A(\lambda) = \lambda^n - \text{Tr}(A)\lambda^{n-1} + c_{n-2}\lambda^{n-2} - \dots + (-1)^n \det(A).$$

In particular:

- $p_A(\lambda)$ is a monic polynomial of degree n (its leading coefficient is 1).
- The coefficient of λ^{n-1} is $-\text{Tr}(A)$.
- The constant term $p_A(0)$ is $(-1)^n \det(A)$.
- The roots of $p_A(\lambda) = 0$ in $\mathbb{K}$ (or an algebraic extension of $\mathbb{K}$) are precisely the eigenvalues of A .

Proof. By definition, the characteristic polynomial of $A = (a_{ij})_{n \times n} \in M_n(\mathbb{K})$ is given by $p_A(\lambda) = \det(\lambda I_n - A)$, which is the determinant of the matrix:

$$\lambda I_n - A = \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}.$$

By the Leibniz formula for determinants, $\det(\lambda I_n - A)$ is expressed as the sum:

$$p_A(\lambda) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n (\lambda I_n - A)_{i, \sigma(i)},$$

where S_n is the symmetric group of permutations on $\{1, 2, \dots, n\}$.

We prove each part of the theorem explicitly:

(i) For the identity permutation $\sigma = \text{id}$, the corresponding term in the Leibniz sum is the product of the main diagonal entries:

$$\prod_{i=1}^n (\lambda - a_{ii}) = (\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn}) = \lambda^n - \left(\sum_{i=1}^n a_{ii} \right) \lambda^{n-1} + \dots$$

For any non-identity permutation $\sigma \neq \text{id}$, $\sigma(i) \neq i$ for at least two indices i . Thus, the product contains at most $n - 2$ diagonal entries of the form $(\lambda - a_{ii})$, meaning its degree in λ is at most $n - 2$.

Therefore, the leading term of $p_A(\lambda)$ comes exclusively from the identity permutation term, giving $1 \cdot \lambda^n = \lambda^n$. Hence, $p_A(\lambda)$ is a monic polynomial of degree n .

(ii) From the Leibniz expansion, non-identity permutations ($\sigma \neq \text{id}$) contribute terms of degree at most $n - 2$ in λ . Consequently, the term involving λ^{n-1} arises solely from expanding the main diagonal product:

$$\prod_{i=1}^n (\lambda - a_{ii}) = \lambda^n - (a_{11} + a_{22} + \dots + a_{nn}) \lambda^{n-1} + \text{lower-degree terms}.$$

Since $\text{Tr}(A) = \sum_{i=1}^n a_{ii}$, the coefficient of λ^{n-1} in $p_A(\lambda)$ is exactly $-\text{Tr}(A)$.

(iii) The constant term of $p_A(\lambda)$ is equal to its value at $\lambda = 0$:

$$p_A(0) = \det(0 \cdot I_n - A) = \det(-A).$$

By the multilinearity of the determinant function, factoring out -1 from each of the n rows yields:

$$\det(-A) = (-1)^n \det(A).$$

Thus, the constant term $p_A(0)$ is $(-1)^n \det(A)$.

(iv) Let $\lambda_0 \in \mathbb{K}$ (or an algebraic extension of $\mathbb{K}$). By definition, λ_0 is an eigenvalue of A if and only if there exists a non-zero eigenvector $v \in \mathbb{K}^n \setminus \{0\}$ satisfying:

$$\begin{aligned} Av = \lambda_0 v &\iff \lambda_0 v - Av = 0 \\ &\iff (\lambda_0 I_n - A)v = 0. \end{aligned}$$

The homogeneous linear system $(\lambda_0 I_n - A)v = 0$ has a non-trivial solution $v \neq 0$ if and only if the coefficient matrix $(\lambda_0 I_n - A)$ is singular (non-invertible), which holds if and only if its determinant vanishes:

$$\det(\lambda_0 I_n - A) = 0 \iff p_A(\lambda_0) = 0.$$

Therefore, λ_0 is a root of $p_A(\lambda) = 0$ if and only if λ_0 is an eigenvalue of A . ■

Theorem 4.2.3 (Similarity Invariance of the Characteristic Polynomial). *Similar matrices have identical characteristic polynomials.*

Proof. Let $A, B \in M_n(\mathbb{K})$ be similar matrices, so $B = P^{-1}AP$ for some non-singular $P \in M_n(\mathbb{K})$. Using determinant multiplicative properties:

$$\begin{aligned} p_B(\lambda) &= \det(\lambda I_n - B) = \det(\lambda I_n - P^{-1}AP) \\ &= \det(P^{-1}(\lambda I_n)P - P^{-1}AP) = \det(P^{-1}(\lambda I_n - A)P) \\ &= \det(P^{-1}) \det(\lambda I_n - A) \det(P) = \det(\lambda I_n - A) = p_A(\lambda). \end{aligned}$$

Thus, $p_B(\lambda) = p_A(\lambda)$, showing that the characteristic polynomial is a similarity invariant. ■

4.2.1 The Minimal Polynomial

While many polynomials $p(x)$ satisfy $p(A) = 0$, there exists a unique polynomial of lowest degree that annihilates A .

Definition 4.2.4 (Minimal Polynomial). The **minimal polynomial** of a square matrix $A \in M_n(\mathbb{K})$, denoted by $m_A(x)$ (or $\mu_A(x)$), is the unique monic polynomial of minimum degree over $\mathbb{K}$ such that:

$$m_A(A) = 0_{n \times n}.$$

Theorem 4.2.5 (Divisibility Property of the Minimal Polynomial). Let $A \in M_n(\mathbb{K})$, and let $m_A(x)$ be its minimal polynomial. A polynomial $p(x) \in \mathbb{K}[x]$ satisfies $p(A) = 0$ if and only if $m_A(x)$ divides $p(x)$ (i.e., $m_A(x) \mid p(x)$).

Proof. Suppose $m_A(x) \mid p(x)$, so $p(x) = q(x)m_A(x)$ for some polynomial $q(x)$. Substituting A :

$$p(A) = q(A)m_A(A) = q(A) \cdot 0 = 0.$$

Suppose $p(A) = 0$. By the polynomial division algorithm, there exist unique polynomials $q(x)$ and $r(x)$ such that:

$$p(x) = q(x)m_A(x) + r(x), \quad \text{where } r(x) = 0 \text{ or } \deg(r) < \deg(m_A).$$

Evaluating at A :

$$p(A) = q(A)m_A(A) + r(A) \implies 0 = q(A) \cdot 0 + r(A) \implies r(A) = 0.$$

If $r(x) \neq 0$, then $r(x)$ would be an annihilating polynomial for A with $\deg(r) < \deg(m_A)$, contradicting the minimality of $\deg(m_A)$. Thus, $r(x) = 0$, proving $m_A(x) \mid p(x)$. ■

Theorem 4.2.6 (Root Correspondence Theorem). Let $A \in M_n(\mathbb{K})$. A scalar $\lambda \in \mathbb{K}$ is a root of the characteristic polynomial $p_A(x)$ if and only if λ is a root of the minimal polynomial $m_A(x)$.

Consequently, $p_A(x)$ and $m_A(x)$ have the same distinct roots (eigenvalues), differing only in the multiplicities of those roots.

Proof. Let λ be a root of $p_A(x)$, so λ is an eigenvalue of A with non-zero eigenvector v ($Av = \lambda v$). Applying $m_A(A) = 0$ to v :

$$0 = m_A(A)v = m_A(\lambda)v.$$

Since $v \neq 0$, $m_A(\lambda) = 0$, so λ is a root of $m_A(x)$.

Conversely, let λ be a root of $m_A(x)$, so $m_A(x) = (x - \lambda)q(x)$ for some polynomial $q(x)$ with $\deg(q) < \deg(m_A)$. Since $m_A(x)$ is minimal, $q(A) \neq 0$. Choose a vector w such that $v = q(A)w \neq 0$. Then:

$$(A - \lambda I_n)v = (A - \lambda I_n)q(A)w = m_A(A)w = 0w = 0.$$

Thus, $Av = \lambda v$ with $v \neq 0$, showing λ is an eigenvalue of A , and hence a root of $p_A(x)$. ■

If the characteristic polynomial of $A \in M_n(\mathbb{K})$ factorizes into linear factors over $\mathbb{K}$ as:

$$p_A(x) = (x - \lambda_1)^{n_1}(x - \lambda_2)^{n_2} \cdots (x - \lambda_k)^{n_k}, \quad \text{where } \sum_{i=1}^k n_i = n,$$

then its minimal polynomial $m_A(x)$ has the factorization structure:

$$m_A(x) = (x - \lambda_1)^{m_1}(x - \lambda_2)^{m_2} \cdots (x - \lambda_k)^{m_k}, \quad \text{where } 1 \leq m_i \leq n_i \text{ for each } i.$$

Remark 4.2.7 (Further Study). A deeper course on canonical forms relates the exponents in the minimal polynomial to Jordan blocks. No knowledge of Jordan form is required here; in this course, minimal polynomials are determined directly by testing possible divisors of the characteristic polynomial.

Example 4.2.8. Find the characteristic polynomial $p_A(x)$ and minimal polynomial $m_A(x)$ for the following matrices:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Solution. For matrix $A = 3I_3$: The characteristic polynomial is:

$$p_A(x) = \det(xI_3 - A) = \det \begin{bmatrix} x-3 & 0 & 0 \\ 0 & x-3 & 0 \\ 0 & 0 & x-3 \end{bmatrix} = (x-3)^3.$$

By the Root Correspondence Theorem, the minimal polynomial must be of the form $m_A(x) = (x-3)^m$ for some $1 \leq m \leq 3$. Testing $m = 1$: $m(A) = A - 3I_3 = 3I_3 - 3I_3 = 0_{3 \times 3}$. Thus, the minimal polynomial is $m_A(x) = x - 3$.

For matrix B : The characteristic polynomial is:

$$p_B(x) = \det(xI_3 - B) = \det \begin{bmatrix} x-3 & -1 & 0 \\ 0 & x-3 & 0 \\ 0 & 0 & x-3 \end{bmatrix} = (x-3)^3.$$

The candidate minimal polynomials are $m_1(x) = x-3$, $m_2(x) = (x-3)^2$, and $m_3(x) = (x-3)^3$.

- Test $m_1(B) = B - 3I_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq 0$.

- Test $m_2(B) = (B - 3I_3)^2$:

$$(B - 3I_3)^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0.$$

Thus, the minimal polynomial for B is $m_B(x) = (x-3)^2$. ■

Example 4.2.9. Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$. Find the characteristic polynomial and the minimal polynomial of A .

Solution. Since A is upper triangular, $p_A(x) = (x-2)^2(x+1)$. Therefore the minimal polynomial must contain the factors $(x-2)$ and $(x+1)$ and must divide $p_A(x)$. The possible candidates are $q_1(x) = (x-2)(x+1)$ and $q_2(x) = (x-2)^2(x+1)$.

Now

$$(A - 2I_3)(A + I_3) = \begin{bmatrix} 0 & 3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq 0,$$

so $q_1(A) \neq 0$. Hence q_1 does not annihilate A . By the Cayley–Hamilton theorem, $p_A(A) = 0$, so $q_2(A) = 0$. Therefore $m_A(x) = (x-2)^2(x+1)$. ■

4.3 Eigenvalues and Eigenvectors as Invariants of Linear Transformations

When a linear operator $T : V \rightarrow V$ on a finite-dimensional vector space V is represented by a matrix, the matrix entries depend on the specific choice of basis for V . However, certain fundamental spectral quantities—namely **eigenvalues**, **eigenspaces**, **characteristic polynomials**, **trace**, and **determinant**—remain invariant under basis changes.

Definition 4.3.1 (Eigenvalue, Eigenvector, and Eigenspace). Let V be a vector space over a field $\mathbb{K}$, and let $T : V \rightarrow V$ be a linear operator.

- (i) A scalar $\lambda \in \mathbb{K}$ is called an **eigenvalue** of T if there exists a non-zero vector $v \in V \setminus \{0\}$ such that: $T(v) = \lambda v$.
- (ii) Any non-zero vector $v \in V \setminus \{0\}$ satisfying $T(v) = \lambda v$ is called an **eigenvector** of T corresponding to λ .
- (iii) The set of all eigenvectors corresponding to λ , together with the zero vector, forms a subspace of V called the **eigenspace** of T corresponding to λ , denoted by $E_\lambda(T)$ or V_λ :

$$E_\lambda(T) = \ker(T - \lambda I) = \{v \in V \mid T(v) = \lambda v\}.$$

For a square matrix $A \in M_n(\mathbb{K})$, these definitions apply to the linear operator $T(x) = Ax$ on $\mathbb{K}^n$, giving $Ax = \lambda x$ and $E_\lambda(A) = \text{null}(A - \lambda I_n)$.

Definition 4.3.2 (Algebraic and Geometric Multiplicity). Let λ be an eigenvalue of $A \in M_n(\mathbb{K})$.

- (i) The **algebraic multiplicity** $m_a(\lambda)$ is the multiplicity of λ as a root of the characteristic polynomial $p_A(x) = \det(xI_n - A)$.
- (ii) The **geometric multiplicity** $m_g(\lambda)$ is the dimension of the eigenspace $E_\lambda(A)$:

$$m_g(\lambda) = \dim(E_\lambda(A)) = \text{nullity}(A - \lambda I_n) = n - \text{rank}(A - \lambda I_n).$$

Theorem 4.3.3 (Fundamental Multiplicity Inequality). For every eigenvalue λ of a matrix $A \in M_n(\mathbb{K})$, the geometric multiplicity is at least 1 and cannot exceed the algebraic multiplicity: $1 \leq m_g(\lambda) \leq m_a(\lambda)$.

4.3.1 Basis Independence and Invariance Under Linear Transformations

Let V be an n -dimensional vector space over $\mathbb{K}$, and let $T : V \rightarrow V$ be a linear operator. If $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_n\}$ are two ordered bases for V , the matrix representations $A = [T]_{\mathcal{B}}$ and $B = [T]_{\mathcal{C}}$ are related by a similarity transformation: $B = P^{-1}AP$, where $P = [I_V]_{\mathcal{B}}^{\mathcal{C}}$ is the non-singular change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$.

Theorem 4.3.4 (Invariance of Eigenvalues under Change of Basis). Let $T : V \rightarrow V$ be a linear operator, and let $A = [T]_{\mathcal{B}}$ and $B = [T]_{\mathcal{C}}$ be matrix representations of T with respect to two different bases $\mathcal{B}$ and $\mathcal{C}$. Then:

- (i) A and B have the same characteristic polynomial: $p_A(\lambda) = p_B(\lambda)$.
- (ii) A and B have the same eigenvalues with identical algebraic and geometric multiplicities.
- (iii) $\text{Tr}(A) = \text{Tr}(B)$ and $\det(A) = \det(B)$.

Consequently, eigenvalues are intrinsic properties of the operator T , independent of the basis chosen to represent T .

Proof. Since $B = P^{-1}AP$ for a non-singular change-of-basis matrix P :

$$\begin{aligned} p_B(\lambda) &= \det(\lambda I_n - B) = \det(\lambda I_n - P^{-1}AP) \\ &= \det(P^{-1}(\lambda I_n - A)P) = \det(P^{-1}) \det(\lambda I_n - A) \det(P) \\ &= \det(\lambda I_n - A) = p_A(\lambda). \end{aligned}$$

Since $p_A(\lambda) = p_B(\lambda)$, A and B share identical roots (eigenvalues) and algebraic multiplicities. Furthermore, since $\text{rank}(P^{-1}MP) = \text{rank}(M)$ for any matrix M :

$$m_g(\lambda)_B = n - \text{rank}(B - \lambda I_n) = n - \text{rank}(P^{-1}(A - \lambda I_n)P) = n - \text{rank}(A - \lambda I_n) = m_g(\lambda)_A.$$

Thus, geometric multiplicities are also preserved. ■

Theorem 4.3.5 (Transformation of Eigenvector Coordinates). *Let $T : V \rightarrow V$ be a linear operator, $A = [T]_B$, $B = [T]_C$, and $B = P^{-1}AP$. If $x \in \mathbb{K}^n$ is an eigenvector of A corresponding to eigenvalue λ ($Ax = \lambda x$), then: $y = P^{-1}x$ is the eigenvector of B corresponding to the same eigenvalue λ ($By = \lambda y$).*

Proof. Substitute $y = P^{-1}x$ into By :

$$By = (P^{-1}AP)(P^{-1}x) = P^{-1}A(P P^{-1})x = P^{-1}Ax = P^{-1}(\lambda x) = \lambda(P^{-1}x) = \lambda y.$$

Since P is non-singular and $x \neq 0$, $y = P^{-1}x \neq 0$. Thus, y is an eigenvector of B with eigenvalue λ . ■

4.3.2 Linear Independence of Eigenvectors

Theorem 4.3.6 (Linear Independence of Eigenvectors for Distinct Eigenvalues). *Let $T : V \rightarrow V$ be a linear operator. If $v_1, v_2, \dots, v_k \in V$ are eigenvectors of T corresponding to **distinct** eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{K}$ (so $\lambda_i \neq \lambda_j$ for $i \neq j$), then the set $\{v_1, v_2, \dots, v_k\}$ is linearly independent.*

Proof. We prove the theorem by induction on k , the number of eigenvectors.

Base Case ($k = 1$): Since v_1 is an eigenvector, $v_1 \neq 0$. The single-element set $\{v_1\}$ is linearly independent.

Inductive Hypothesis: Assume that any set of $k - 1$ eigenvectors corresponding to distinct eigenvalues is linearly independent.

Inductive Step: Let $v_1, v_2, \dots, v_k$ be eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Consider the vector relation:

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0. \tag{1}$$

Apply the linear operator T to both sides of equation (1):

$$\begin{aligned} T(c_1 v_1 + c_2 v_2 + \dots + c_k v_k) &= T(0) \\ c_1 T(v_1) + c_2 T(v_2) + \dots + c_k T(v_k) &= 0 \\ c_1 \lambda_1 v_1 + c_2 \lambda_2 v_2 + \dots + c_k \lambda_k v_k &= 0. \end{aligned} \tag{2}$$

Multiply equation (1) by the scalar λ_k :

$$c_1 \lambda_k v_1 + c_2 \lambda_k v_2 + \dots + c_k \lambda_k v_k = 0. \tag{3}$$

Subtract equation (3) from equation (2):

$$c_1(\lambda_1 - \lambda_k)v_1 + c_2(\lambda_2 - \lambda_k)v_2 + \dots + c_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1} + 0v_k = 0.$$

This is a linear combination of the $k - 1$ eigenvectors $\{v_1, \dots, v_{k-1}\}$. By the inductive hypothesis, $\{v_1, \dots, v_{k-1}\}$ is linearly independent, so every coefficient must be zero: $c_i(\lambda_i - \lambda_k) = 0$ for all $1 \leq i \leq k - 1$. Since all eigenvalues are distinct, $\lambda_i - \lambda_k \neq 0$ for $i < k$, which forces $c_i = 0$ for $1 \leq i \leq k - 1$.

Substituting $c_1 = c_2 = \dots = c_{k-1} = 0$ back into equation (1) leaves $c_k v_k = 0$. Since $v_k \neq 0$, $c_k = 0$.

Thus, $c_1 = c_2 = \dots = c_k = 0$, proving $\{v_1, v_2, \dots, v_k\}$ is linearly independent. $\blacksquare$

Example 4.3.7. Find the eigenvalues, algebraic multiplicities, geometric multiplicities, and eigenspaces for the matrix:

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \in M_3(\mathbb{R}).$$

Solution. First, compute the characteristic polynomial $p_A(\lambda) = \det(\lambda I_3 - A)$:

$$\begin{aligned} p_A(\lambda) &= \det \begin{bmatrix} \lambda - 4 & 0 & -1 \\ 2 & \lambda - 1 & 0 \\ 2 & 0 & \lambda - 1 \end{bmatrix} \\ &= (\lambda - 1) \det \begin{bmatrix} \lambda - 4 & -1 \\ 2 & \lambda - 1 \end{bmatrix} \\ &= (\lambda - 1) ((\lambda - 4)(\lambda - 1) - (-2)) \\ &= (\lambda - 1)(\lambda^2 - 5\lambda + 6) = (\lambda - 1)(\lambda - 2)(\lambda - 3). \end{aligned}$$

The eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$. Each eigenvalue has algebraic multiplicity $m_a = 1$. Since $1 \leq m_g \leq m_a = 1$, each eigenvalue also has geometric multiplicity $m_g = 1$.

We now compute the eigenspace corresponding to each eigenvalue.

For $\lambda_1 = 1$: Solve $(A - 1I_3)x = 0$:

$$\begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies x_1 = 0, \quad x_3 = 0, \quad x_2 \text{ is free.}$$

$$E_1(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

For $\lambda_2 = 2$: Solve $(A - 2I_3)x = 0$:

$$\begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies x_3 = -2x_1, \quad x_2 = -2x_1.$$

$$E_2(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \right\}.$$

For $\lambda_3 = 3$: Solve $(A - 3I_3)x = 0$:

$$\begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies x_3 = -x_1, \quad x_2 = -x_1.$$

$$E_3(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} \right\}.$$

Since $\lambda_1, \lambda_2, \lambda_3$ are distinct, the three basis vectors are linearly independent and form an eigen-vector basis for $\mathbb{R}^3$. ■

Example 4.3.8. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Compute $B = P^{-1}AP$, verify that $\det(xI_2 - A) = \det(xI_2 - B)$, and show how the eigenvectors of A transform into eigenvectors of B .

Solution. First, compute P^{-1} . Since $\det(P) = (1)(-1) - (1)(1) = -2$:

$$P^{-1} = -\frac{1}{2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

Now compute $B = P^{-1}AP$:

$$\begin{aligned} AP &= \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 5 & 1 \end{bmatrix}, \\ B &= P^{-1}(AP) = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -1 & -1 \end{bmatrix}. \end{aligned}$$

The characteristic polynomials are

$$\begin{aligned} p_A(x) &= \det \begin{bmatrix} x-1 & -2 \\ -3 & x-2 \end{bmatrix} = (x-1)(x-2) - 6 = x^2 - 3x - 4 = (x-4)(x+1), \\ p_B(x) &= \det \begin{bmatrix} x-4 & 0 \\ 1 & x+1 \end{bmatrix} = (x-4)(x+1) = x^2 - 3x - 4. \end{aligned}$$

Thus, $p_A(x) = p_B(x)$, and both matrices share eigenvalues $\lambda_1 = 4$ and $\lambda_2 = -1$.

For $\lambda_1 = 4$, an eigenvector x_1 of A satisfies $(A - 4I_2)x_1 = 0$:

$$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

The corresponding eigenvector y_1 of B is given by $y_1 = P^{-1}x_1$:

$$y_1 = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5/2 \\ -1/2 \end{bmatrix}.$$

Verifying $By_1 = 4y_1$:

$$By_1 = \begin{bmatrix} 4 & 0 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 5/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 10 \\ -2 \end{bmatrix} = 4 \begin{bmatrix} 5/2 \\ -1/2 \end{bmatrix} = 4y_1.$$

This confirms that $y_1 = P^{-1}x_1$ is an eigenvector of B with eigenvalue $\lambda_1 = 4$. ■

4.4 The Cayley–Hamilton Theorem and Its Computational Applications

The **Cayley–Hamilton Theorem** is one of the foundational results in linear algebra. It asserts that every square matrix satisfies its own characteristic equation, converting higher-degree matrix power calculations and matrix inversions into manageable lower-degree polynomial evaluations.

Theorem 4.4.1 (Cayley–Hamilton Theorem). *Let $A \in M_n(\mathbb{K})$ be a square matrix over a field $\mathbb{K}$, and let $p_A(\lambda) = \det(\lambda I_n - A)$ be its characteristic polynomial, expanded as:*

$$p_A(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0.$$

Then the matrix A satisfies its own characteristic equation:

$$p_A(A) = A^n + c_{n-1}A^{n-1} + \dots + c_1A + c_0I_n = 0_{n \times n}.$$

Proof. Let $C(\lambda) = \lambda I_n - A$ be the characteristic matrix of A . The classical adjugate (adjoint) matrix $B(\lambda) = \text{adj}(\lambda I_n - A)$ is a matrix whose entries are cofactors of $\lambda I_n - A$, which are polynomials in λ of degree at most $n - 1$.

Therefore, $B(\lambda)$ can be uniquely expanded as a matrix polynomial in λ with constant coefficient matrices $B_0, B_1, \dots, B_{n-1} \in M_n(\mathbb{K})$:

$$B(\lambda) = B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \dots + B_1\lambda + B_0.$$

By the fundamental property of the adjugate matrix ($M \cdot \text{adj}(M) = \det(M)I_n$):

$$(\lambda I_n - A)B(\lambda) = \det(\lambda I_n - A)I_n = p_A(\lambda)I_n.$$

Substituting the expansion of $B(\lambda)$ and $p_A(\lambda)$:

$$(\lambda I_n - A)(B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \dots + B_0) = (\lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0)I_n.$$

Expanding the left-hand side:

$$B_{n-1}\lambda^n + (B_{n-2} - AB_{n-1})\lambda^{n-1} + (B_{n-3} - AB_{n-2})\lambda^{n-2} + \dots + (B_0 - AB_1)\lambda - AB_0.$$

Equating coefficients of identical powers of λ on both sides:

$$\begin{aligned} B_{n-1} &= I_n, \\ B_{n-2} - AB_{n-1} &= c_{n-1}I_n, \\ B_{n-3} - AB_{n-2} &= c_{n-2}I_n, \\ &\vdots \\ B_0 - AB_1 &= c_1I_n, \\ -AB_0 &= c_0I_n. \end{aligned}$$

Left-multiply the first equation by A^n , the second by A^{n-1} , the third by A^{n-2} , $\dots$, the k -th by A^{n-k+1} , and the last by $A^0 = I_n$:

$$\begin{aligned} A^n B_{n-1} &= A^n, \\ A^{n-1} B_{n-2} - A^n B_{n-1} &= c_{n-1} A^{n-1}, \\ A^{n-2} B_{n-3} - A^{n-1} B_{n-2} &= c_{n-2} A^{n-2}, \\ &\vdots \\ AB_0 - A^2 B_1 &= c_1 A, \\ -AB_0 &= c_0 I_n. \end{aligned}$$

Summing all left-hand sides yields a telescoping sum where every intermediate matrix term cancels out completely:

$$0_{n \times n} = A^n + c_{n-1}A^{n-1} + c_{n-2}A^{n-2} + \dots + c_1A + c_0I_n = p_A(A).$$

Hence, $p_A(A) = 0_{n \times n}$. ■

The Cayley–Hamilton theorem provides powerful techniques for three primary computational tasks:

Matrix inversion. If $A \in M_n(\mathbb{K})$ is non-singular, then $\det(A) \neq 0$, which implies $c_0 = (-1)^n \det(A) \neq 0$.

By the Cayley–Hamilton Theorem:

$$A^n + c_{n-1}A^{n-1} + \dots + c_1A + c_0I_n = 0 \implies -c_0I_n = A(A^{n-1} + c_{n-1}A^{n-2} + \dots + c_1I_n).$$

Multiplying by $-\frac{1}{c_0}$ gives an explicit formula for the inverse A^{-1} :

$$A^{-1} = -\frac{1}{c_0} (A^{n-1} + c_{n-1}A^{n-2} + \dots + c_2A + c_1I_n).$$

Reduction of higher matrix powers. Any matrix power A^m ($m \geq n$) can be expressed as a linear combination of $I_n, A, A^2, \dots, A^{n-1}$.

By the polynomial division algorithm, divide x^m by the characteristic polynomial $p_A(x)$: $x^m = q(x)p_A(x) + r(x)$, where $\deg(r) < n$. Let $r(x) = \alpha_{n-1}x^{n-1} + \dots + \alpha_1x + \alpha_0$. Substituting A :

$$A^m = q(A)p_A(A) + r(A).$$

Since $p_A(A) = 0$, this simplifies to: $A^m = r(A) = \alpha_{n-1}A^{n-1} + \dots + \alpha_1A + \alpha_0I_n$.

Evaluation of matrix polynomials and functions. For any high-degree polynomial $f(x)$, divide $f(x)$ by $p_A(x)$ to get $f(x) = q(x)p_A(x) + r(x)$, yielding $f(A) = r(A)$.

Example 4.4.2. Verify the Cayley–Hamilton Theorem for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, and use it to compute A^{-1} and A^4 .

Solution. Compute the characteristic polynomial $p_A(\lambda) = \det(\lambda I_2 - A)$:

$$p_A(\lambda) = \det \begin{bmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 4 \end{bmatrix} = (\lambda - 1)(\lambda - 4) - 6 = \lambda^2 - 5\lambda - 2.$$

To verify $p_A(A) = A^2 - 5A - 2I_2 = 0$:

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}.$$

$$A^2 - 5A - 2I_2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This verifies the Cayley–Hamilton Theorem.

To compute A^{-1} , rearrange $A^2 - 5A - 2I_2 = 0 \implies 2I_2 = A^2 - 5A = A(A - 5I_2)$:

$$A^{-1} = \frac{1}{2}(A - 5I_2) = \frac{1}{2} \left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}.$$

To compute A^4 , rearrange $A^2 = 5A + 2I_2$:

$$A^3 = A \cdot A^2 = A(5A + 2I_2) = 5A^2 + 2A = 5(5A + 2I_2) + 2A = 27A + 10I_2,$$

$$A^4 = A \cdot A^3 = A(27A + 10I_2) = 27A^2 + 10A = 27(5A + 2I_2) + 10A = 145A + 54I_2.$$

Substituting A and I_2 :

$$A^4 = 145 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + 54 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 145 + 54 & 290 \\ 435 & 580 + 54 \end{bmatrix} = \begin{bmatrix} 199 & 290 \\ 435 & 634 \end{bmatrix}.$$



Example 4.4.3. Verify the Cayley–Hamilton Theorem for $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, and compute A^{-1} .

Solution. Compute $p_A(\lambda) = \det(\lambda I_3 - A)$:

$$p_A(\lambda) = \lambda^3 - \text{Tr}(A)\lambda^2 + c_1\lambda - \det(A).$$

- $\text{Tr}(A) = 2 + 2 + 2 = 6$.
- $c_1 = \det \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} + \det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} + \det \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = 3 + 3 + 3 = 9$.
- $\det(A) = 2(4 - 1) - (-1)(-2 + 1) + 1(1 - 2) = 6 - 1 - 1 = 4$.

Thus, $p_A(\lambda) = \lambda^3 - 6\lambda^2 + 9\lambda - 4$.

We compute powers of A :

$$A^2 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}.$$

Evaluating $p_A(A) = A^3 - 6A^2 + 9A - 4I_3$:

$$\begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 0_{3 \times 3}.$$

This verifies the theorem.

To compute A^{-1} , use $A^3 - 6A^2 + 9A - 4I_3 = 0 \implies 4I_3 = A(A^2 - 6A + 9I_3)$: $A^{-1} = \frac{1}{4}(A^2 - 6A + 9I_3)$.

$$A^2 - 6A + 9I_3 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

Thus:

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

■

Example 4.4.4. Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Evaluate $f(A) = A^5 - 4A^4 + 3A^3 + 2A^2 - 5A + 4I_2$.

Solution. Compute the characteristic polynomial of A :

$$p_A(\lambda) = \det \begin{bmatrix} \lambda - 2 & -1 \\ -1 & \lambda - 2 \end{bmatrix} = (\lambda - 2)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

By the Cayley–Hamilton Theorem, $A^2 - 4A + 3I_2 = 0$.

Perform polynomial division of $f(x) = x^5 - 4x^4 + 3x^3 + 2x^2 - 5x + 4$ by $p_A(x) = x^2 - 4x + 3$:

$$\begin{aligned} f(x) &= x^3(x^2 - 4x + 3) + 2(x^2 - 4x + 3) + 3x - 2 \\ &= (x^3 + 2)(x^2 - 4x + 3) + (3x - 2). \end{aligned}$$

Substituting A into $f(A)$:

$$f(A) = (A^3 + 2I_2) \underbrace{(A^2 - 4A + 3I_2)}_{=0} + (3A - 2I_2) = 3A - 2I_2.$$

Evaluating $3A - 2I_2$:

$$f(A) = 3 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 6-2 & 3 \\ 3 & 6-2 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}.$$

■

4.5 Inner Product Spaces on $\mathbb{R}^n$ and $\mathbb{C}^n$, Norms, and Orthogonality

To introduce geometric notions such as length, distance, angle, and orthogonality into $\mathbb{R}^n$ and $\mathbb{C}^n$, we equip these spaces of column vectors with an **inner product structure**.

Definition 4.5.1 (Inner Product on $\mathbb{K}^n$). Let $\mathbb{K}^n$ denote the space of column vectors with entries in $\mathbb{K}$ (where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$). An **inner product** on $\mathbb{K}^n$ is a function $\langle \cdot, \cdot \rangle : \mathbb{K}^n \times \mathbb{K}^n \rightarrow \mathbb{K}$ that assigns a scalar $\langle x, y \rangle \in \mathbb{K}$ to each pair of vectors $x, y \in \mathbb{K}^n$, satisfying the following axioms for all vectors $x, y, z \in \mathbb{K}^n$ and all scalars $\alpha, \beta \in \mathbb{K}$:

(i) **Conjugate Symmetry:** $\langle x, y \rangle = \overline{\langle y, x \rangle}$. (When $\mathbb{K} = \mathbb{R}$, this simplifies to standard symmetry: $\langle x, y \rangle = \langle y, x \rangle$).

(ii) **Linearity in the Second Argument:**

$$\langle x, \alpha y + \beta z \rangle = \alpha \langle x, y \rangle + \beta \langle x, z \rangle.$$

(iii) **Conjugate Linearity in the First Argument:**

$$\langle \alpha x + \beta y, z \rangle = \bar{\alpha} \langle x, z \rangle + \bar{\beta} \langle y, z \rangle.$$

(iv) **Positive Definiteness:**

$$\langle x, x \rangle \geq 0 \quad \text{for all } x \in \mathbb{K}^n, \quad \text{and} \quad \langle x, x \rangle = 0 \iff x = 0.$$

The space $\mathbb{K}^n$ equipped with an inner product $\langle \cdot, \cdot \rangle$ is called an **inner product space**.

Example 4.5.2. On $\mathbb{R}^n$, the standard inner product (dot product) of $x = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T$ and $y = \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix}^T$ is:

$$\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i.$$

For $x = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $y = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^3$: $\langle x, y \rangle = (1)(4) + (2)(-1) + (-3)(2) = 4 - 2 - 6 = -4$.

Example 4.5.3. On $\mathbb{C}^n$, the standard complex inner product of $x = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T$ and $y = \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix}^T$ is:

$$\langle x, y \rangle = x^* y = \sum_{i=1}^n \bar{x}_i y_i.$$

For $x = \begin{bmatrix} 1+i \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 3i \\ 1-i \end{bmatrix}$ in $\mathbb{C}^2$:

$$\begin{aligned}\langle x, y \rangle &= \overline{(1+i)}(3i) + \overline{2}(1-i) \\ &= (1-i)(3i) + 2(1-i) \\ &= (3i - 3i^2) + (2 - 2i) = (3i + 3) + 2 - 2i = 5 + i.\end{aligned}$$

Example 4.5.4. Let $A \in M_n(\mathbb{R})$ be a real symmetric, positive-definite matrix (so $A^T = A$ and $x^T A x > 0$ for all $x \neq 0$). The function defined by: $\langle x, y \rangle_A = x^T A y$ is a valid inner product on $\mathbb{R}^n$.

For example, choosing the diagonal matrix $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ gives the weighted inner product on $\mathbb{R}^2$: $\langle x, y \rangle_A = 2x_1y_1 + 3x_2y_2$. For $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$: $\langle x, y \rangle_A = 2(1)(4) + 3(2)(1) = 8 + 6 = 14$.

Example 4.5.5. For real $m \times n$ matrices $A, B \in M_{m \times n}(\mathbb{R})$, the **Frobenius inner product** is defined using the trace operator:

$$\langle A, B \rangle_F = \text{Tr}(A^T B) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij}.$$

For $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 4 & 1 \end{bmatrix}$ in $M_2(\mathbb{R})$: $\langle A, B \rangle_F = 1(2) + 2(-1) + 0(4) + 3(1) = 2 - 2 + 0 + 3 = 3$.

Example 4.5.6. For complex $m \times n$ matrices $A, B \in M_{m \times n}(\mathbb{C})$, the **complex Frobenius inner product** is defined by:

$$\langle A, B \rangle_F = \text{Tr}(A^* B) = \sum_{i=1}^m \sum_{j=1}^n \bar{a}_{ij} b_{ij}.$$

For $A = \begin{bmatrix} i & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2i & 0 \\ 1 & 1+i \end{bmatrix}$ in $M_2(\mathbb{C})$:

$$\begin{aligned}\langle A, B \rangle_F &= \bar{i}(2i) + \bar{1}(0) + \bar{0}(1) + \bar{2}(1+i) \\ &= (-i)(2i) + 0 + 0 + 2(1+i) \\ &= -2i^2 + 2 + 2i = 2 + 2 + 2i = 4 + 2i.\end{aligned}$$

Example 4.5.7. Let $C[a, b]$ be the space of continuous real-valued functions defined on the closed interval $[a, b]$. The standard inner product on $C[a, b]$ is given by integration: $\langle f, g \rangle = \int_a^b f(x)g(x) dx$. For $f(x) = x$ and $g(x) = x^2$ on the interval $[0, 1]$:

$$\langle f, g \rangle = \int_0^1 (x)(x^2) dx = \int_0^1 x^3 dx = \left[\frac{x^4}{4} \right]_0^1 = \frac{1}{4}.$$

Example 4.5.8. For complex-valued continuous functions $f, g : [a, b] \rightarrow \mathbb{C}$, the inner product is given by: $\langle f, g \rangle = \int_a^b \overline{f(x)}g(x) dx$. For $f(x) = e^{ix}$ and $g(x) = e^{2ix}$ on the interval $[0, 2\pi]$:

$$\begin{aligned}\langle f, g \rangle &= \int_0^{2\pi} \overline{e^{ix}} e^{2ix} dx = \int_0^{2\pi} e^{-ix} e^{2ix} dx \\ &= \int_0^{2\pi} e^{ix} dx = \left[\frac{e^{ix}}{i} \right]_0^{2\pi} = \frac{e^{2\pi i} - e^0}{i} = \frac{1 - 1}{i} = 0.\end{aligned}$$

Thus, e^{ix} and e^{2ix} are orthogonal in this function space.

4.5.1 Standard Inner Products on $\mathbb{R}^n$ and $\mathbb{C}^n$

Definition 4.5.9 (Real Space $\mathbb{R}^n$). The **standard inner product** (or Euclidean dot product) on $\mathbb{R}^n$ is a function $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined for vectors $x = [x_1 \ \dots \ x_n]^T$ and $y = [y_1 \ \dots \ y_n]^T$ by:

$$\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i = x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$$

Definition 4.5.10 (Complex Space $\mathbb{C}^n$). The **standard inner product** on $\mathbb{C}^n$ is a function $\langle \cdot, \cdot \rangle : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ defined for vectors $x = [x_1 \ \dots \ x_n]^T$ and $y = [y_1 \ \dots \ y_n]^T$ by:

$$\langle x, y \rangle = x^* y = (\bar{x})^T y = \sum_{i=1}^n \bar{x}_i y_i = \bar{x}_1 y_1 + \bar{x}_2 y_2 + \dots + \bar{x}_n y_n.$$

4.5.2 Induced Norms and Fundamental Inequalities

Definition 4.5.11 (Induced Vector Norm). The **norm** (or length) of a vector $x \in \mathbb{K}^n$ induced by the standard inner product is the non-negative real number $\|x\|$ defined by:

$$\|x\| = \sqrt{\langle x, x \rangle} = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2}.$$

A vector $u \in \mathbb{K}^n$ with $\|u\| = 1$ is called a **unit vector** (or **normalized vector**).

Theorem 4.5.12 (Cauchy–Schwarz Inequality). For any two vectors $x, y \in \mathbb{K}^n$: $|\langle x, y \rangle| \leq \|x\| \|y\|$, with equality holding if and only if x and y are linearly dependent.

Proof. If $y = 0$, then $\langle x, 0 \rangle = 0$ and $\|x\| \|0\| = 0$, so equality holds trivially.

Assume $y \neq 0$, so $\|y\| > 0$. For any scalar $t \in \mathbb{K}$, positive definiteness implies:

$$0 \leq \|x - ty\|^2 = \langle x - ty, x - ty \rangle = \langle x, x \rangle - \bar{t} \langle y, x \rangle - t \langle x, y \rangle + |t|^2 \langle y, y \rangle.$$

Choose the specific scalar $t = \frac{\langle y, x \rangle}{\langle y, y \rangle} = \frac{\overline{\langle x, y \rangle}}{\|y\|^2}$. Substituting t into the inequality:

$$\begin{aligned} 0 &\leq \|x\|^2 - \frac{\langle x, y \rangle}{\|y\|^2} \langle y, x \rangle - \frac{\overline{\langle x, y \rangle}}{\|y\|^2} \langle x, y \rangle + \frac{|\langle x, y \rangle|^2}{\|y\|^4} \|y\|^2 \\ &= \|x\|^2 - \frac{|\langle x, y \rangle|^2}{\|y\|^2} - \frac{|\langle x, y \rangle|^2}{\|y\|^2} + \frac{|\langle x, y \rangle|^2}{\|y\|^2} \\ &= \|x\|^2 - \frac{|\langle x, y \rangle|^2}{\|y\|^2}. \end{aligned}$$

Rearranging terms yields $|\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$. Taking the positive square root gives: $|\langle x, y \rangle| \leq \|x\| \|y\|$. Equality holds if and only if $\|x - ty\|^2 = 0 \iff x - ty = 0 \iff x = ty$, meaning x and y are linearly dependent. ■

Theorem 4.5.13 (Properties of the Vector Norm). For all $x, y \in \mathbb{K}^n$ and $\alpha \in \mathbb{K}$:

(i) $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$.

(ii) **Absolute Homogeneity**: $\|\alpha x\| = |\alpha| \|x\|$.

(iii) **Triangle Inequality**: $\|x + y\| \leq \|x\| + \|y\|$.

Proof. We prove property (iii) using the Cauchy–Schwarz Inequality:

$$\begin{aligned}
 \|x + y\|^2 &= \langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\
 &= \|x\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} + \|y\|^2 \\
 &= \|x\|^2 + 2\operatorname{Re}(\langle x, y \rangle) + \|y\|^2 \\
 &\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \quad (\text{since } \operatorname{Re}(z) \leq |z|) \\
 &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \quad (\text{by Cauchy–Schwarz}) \\
 &= (\|x\| + \|y\|)^2.
 \end{aligned}$$

Taking square roots yields $\|x + y\| \leq \|x\| + \|y\|$. ■

4.5.3 Orthogonality, Pythagorean Theorem, and Parallelogram Law

Definition 4.5.14 (Orthogonality). Two vectors $x, y \in \mathbb{K}^n$ are said to be **orthogonal** (denoted $x \perp y$) if their inner product is zero: $x \perp y \iff \langle x, y \rangle = 0$.

Theorem 4.5.15 (Pythagorean Theorem in $\mathbb{K}^n$). If $x, y \in \mathbb{K}^n$ are orthogonal vectors ($x \perp y$), then:

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

Proof. Expanding $\|x + y\|^2$:

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2.$$

Since $x \perp y$, $\langle x, y \rangle = 0$ and $\langle y, x \rangle = \overline{\langle x, y \rangle} = 0$. Thus:

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2. \quad \text{■}$$

Theorem 4.5.16 (Parallelogram Law). For any two vectors $x, y \in \mathbb{K}^n$:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Proof. Expand both norm square terms:

$$\begin{aligned}
 \|x + y\|^2 &= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2, \\
 \|x - y\|^2 &= \|x\|^2 - \langle x, y \rangle - \langle y, x \rangle + \|y\|^2.
 \end{aligned}$$

Adding the two equations cancels the cross-terms $\langle x, y \rangle$ and $\langle y, x \rangle$:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2. \quad \text{■}$$

Example 4.5.17. Let $x = \begin{bmatrix} 1+i \\ 2 \\ -i \end{bmatrix}$ and $y = \begin{bmatrix} 2i \\ 1-i \\ 3 \end{bmatrix}$ in $\mathbb{C}^3$. Compute $\langle x, y \rangle$, $\|x\|$, $\|y\|$, and verify $\langle y, x \rangle = \overline{\langle x, y \rangle}$.

Solution. Using the standard complex inner product $\langle x, y \rangle = \sum_{j=1}^3 \bar{x}_j y_j$:

$$\begin{aligned}
 \langle x, y \rangle &= \overline{(1+i)}(2i) + \overline{(2)}(1-i) + \overline{(-i)}(3) \\
 &= (1-i)(2i) + 2(1-i) + (i)(3) \\
 &= (2i - 2i^2) + (2 - 2i) + 3i = (2i + 2) + 2 - 2i + 3i = 4 + 3i.
 \end{aligned}$$

Now compute $\langle y, x \rangle = \sum_{j=1}^3 \bar{y}_j x_j$:

$$\begin{aligned}\langle y, x \rangle &= \overline{(2i)}(1+i) + \overline{(1-i)}(2) + \overline{(3)}(-i) \\ &= (-2i)(1+i) + (1+i)(2) + 3(-i) \\ &= (-2i - 2i^2) + 2 + 2i - 3i = (-2i + 2) + 2 - i = 4 - 3i.\end{aligned}$$

Notice that $\overline{\langle x, y \rangle} = \overline{4 + 3i} = 4 - 3i = \langle y, x \rangle$, confirming conjugate symmetry.

Now compute the norms:

$$\|x\| = \sqrt{|1+i|^2 + |2|^2 + |-i|^2} = \sqrt{(1^2 + 1^2) + 4 + 1^2} = \sqrt{2 + 4 + 1} = \sqrt{7}.$$

$$\|y\| = \sqrt{|2i|^2 + |1-i|^2 + |3|^2} = \sqrt{2^2 + (1^2 + (-1)^2) + 9} = \sqrt{4 + 2 + 9} = \sqrt{15}.$$

■

Example 4.5.18. Using $x = \begin{bmatrix} 1+i \\ 2 \\ -i \end{bmatrix}$ and $y = \begin{bmatrix} 2i \\ 1-i \\ 3 \end{bmatrix}$ from the previous example, verify the Cauchy-Schwarz Inequality $|\langle x, y \rangle| \leq \|x\|\|y\|$.

Solution. From the previous calculations:

$$\langle x, y \rangle = 4 + 3i \implies |\langle x, y \rangle| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5.$$

$$\|x\| = \sqrt{7}, \quad \|y\| = \sqrt{15} \implies \|x\|\|y\| = \sqrt{7 \times 15} = \sqrt{105} \approx 10.246.$$

Comparing the values: $5 \leq \sqrt{105} \implies |\langle x, y \rangle| \leq \|x\|\|y\|$. This verifies the Cauchy-Schwarz Inequality. ■

Example 4.5.19. Let $x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$. Verify the Parallelogram Law $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$.

Solution. Compute the norms of x and y : $\|x\|^2 = 3^2 + 1^2 = 10 \implies 2\|x\|^2 = 20$, $\|y\|^2 = 1^2 + 2^2 = 5 \implies 2\|y\|^2 = 10$. Thus, the right-hand side is $2\|x\|^2 + 2\|y\|^2 = 20 + 10 = 30$.

Now compute $x+y$ and $x-y$:

$$x+y = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \implies \|x+y\|^2 = 4^2 + 3^2 = 25,$$

$$x-y = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \implies \|x-y\|^2 = 2^2 + (-1)^2 = 5.$$

The left-hand side is $\|x+y\|^2 + \|x-y\|^2 = 25 + 5 = 30$.

Since $30 = 30$, the Parallelogram Law holds. ■

4.6 Orthonormal Sets and Geometric Interpretation of Length and Angles

Orthonormal sets represent the ideal coordinate systems in $\mathbb{R}^n$ and $\mathbb{C}^n$. They dramatically simplify vector decompositions, projection calculations, and geometric analysis of length and angles.

4.6.1 Orthogonal and Orthonormal Sets

Definition 4.6.1 (Orthogonal and Orthonormal Sets). Let $S = \{u_1, u_2, \dots, u_k\}$ be a set of column vectors in $\mathbb{K}^n$ ($\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$).

- (i) S is called an **orthogonal set** if every pair of distinct vectors in S is orthogonal: $\langle u_i, u_j \rangle = 0$ for all $i \neq j$.
- (ii) S is called an **orthonormal set** if it is an orthogonal set and every vector in S is a unit vector ($\|u_i\| = 1$). Using the Kronecker delta δ_{ij} :

$$\langle u_i, u_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Theorem 4.6.2 (Linear Independence of Orthogonal Sets). *Every orthogonal set of non-zero vectors $S = \{u_1, u_2, \dots, u_k\} \subset \mathbb{K}^n \setminus \{0\}$ is linearly independent.*

Proof. Suppose there exist scalars $c_1, c_2, \dots, c_k \in \mathbb{K}$ such that: $c_1 u_1 + c_2 u_2 + \dots + c_k u_k = 0$. To isolate the coefficient c_i , take the inner product of both sides with the i -th vector u_i :

$$\left\langle u_i, \sum_{j=1}^k c_j u_j \right\rangle = \langle u_i, 0 \rangle \implies \sum_{j=1}^k c_j \langle u_i, u_j \rangle = 0.$$

Since $\langle u_i, u_j \rangle = 0$ for all $j \neq i$, the sum collapses to a single term: $c_i \langle u_i, u_i \rangle = 0 \implies c_i \|u_i\|^2 = 0$. Since $u_i \neq 0$, $\|u_i\|^2 > 0$, forcing $c_i = 0$. Since this holds for every $i = 1, 2, \dots, k$, S is linearly independent. ■

4.6.2 Orthonormal Bases and Fourier Expansion

Remark 4.6.3 (Further Study). The syllabus requires orthonormal sets and the geometric interpretation of length and angles. Fourier expansion is included here as a useful extension showing how orthonormal bases simplify vector representation.

Definition 4.6.4 (Orthonormal Basis). An **orthonormal basis** (ONB) for $\mathbb{K}^n$ is a basis $\mathcal{U} = \{u_1, u_2, \dots, u_n\}$ that is an orthonormal set.

Theorem 4.6.5 (Fourier Expansion Theorem). *Let $\mathcal{U} = \{u_1, u_2, \dots, u_n\}$ be an orthonormal basis for $\mathbb{K}^n$. Then every vector $x \in \mathbb{K}^n$ can be expressed uniquely as:*

$$x = \sum_{i=1}^n \langle u_i, x \rangle u_i = \langle u_1, x \rangle u_1 + \langle u_2, x \rangle u_2 + \dots + \langle u_n, x \rangle u_n.$$

The scalar coefficients $c_i = \langle u_i, x \rangle$ are called the **Fourier coefficients** of x with respect to $\mathcal{U}$.

Proof. Since $\mathcal{U}$ is a basis, there exist unique scalars $c_1, \dots, c_n$ such that $x = \sum_{j=1}^n c_j u_j$. Taking the inner product of u_i with x :

$$\langle u_i, x \rangle = \left\langle u_i, \sum_{j=1}^n c_j u_j \right\rangle = \sum_{j=1}^n c_j \langle u_i, u_j \rangle = \sum_{j=1}^n c_j \delta_{ij} = c_i.$$

Substituting $c_i = \langle u_i, x \rangle$ back yields the Fourier expansion formula. ■

Theorem 4.6.6 (Parseval's Identity). *Let $\{u_1, u_2, \dots, u_n\}$ be an orthonormal basis for $\mathbb{K}^n$. For any vector $x \in \mathbb{K}^n$, the norm square of x is the sum of the absolute squares of its Fourier coefficients:*

$$\|x\|^2 = \sum_{i=1}^n |\langle u_i, x \rangle|^2.$$

Proof. Substitute the Fourier expansion $x = \sum_{i=1}^n c_i u_i$ into $\|x\|^2 = \langle x, x \rangle$:

$$\|x\|^2 = \left\langle \sum_{i=1}^n c_i u_i, \sum_{j=1}^n c_j u_j \right\rangle = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i c_j \langle u_i, u_j \rangle = \sum_{i=1}^n \bar{c}_i c_i (1) = \sum_{i=1}^n |c_i|^2.$$

Since $c_i = \langle u_i, x \rangle$, $\|x\|^2 = \sum_{i=1}^n |\langle u_i, x \rangle|^2$. ■

4.6.3 The Gram–Schmidt Orthogonalization Process

Remark 4.6.7 (Further Study). Gram–Schmidt is included as an enrichment topic. It provides a systematic way to construct orthonormal sets from linearly independent vectors.

The **Gram–Schmidt process** is an algorithmic procedure that transforms any linearly independent set of column vectors into an orthonormal set spanning the same set of linear combinations.

Theorem 4.6.8 (Gram–Schmidt Orthogonalization Theorem). *Let $\{v_1, v_2, \dots, v_k\}$ be a linearly independent set of vectors in $\mathbb{K}^n$. Then there exists an orthonormal set $\{u_1, u_2, \dots, u_k\}$ such that for every $1 \leq m \leq k$:*

$$\text{span}\{u_1, u_2, \dots, u_m\} = \text{span}\{v_1, v_2, \dots, v_m\}.$$

Construction of the Algorithm. We construct an orthogonal set $\{w_1, w_2, \dots, w_k\}$ sequentially, and then normalize each vector to obtain $u_i = \frac{w_i}{\|w_i\|}$:

First Vector: Set $w_1 = v_1$, and normalize to get $u_1 = \frac{w_1}{\|w_1\|}$.

Second Vector: Subtract the orthogonal projection of v_2 onto u_1 : $w_2 = v_2 - \langle u_1, v_2 \rangle u_1$. Then normalize to get $u_2 = \frac{w_2}{\|w_2\|}$.

General m -th Vector: Having constructed orthonormal vectors $\{u_1, \dots, u_{m-1}\}$, subtract the orthogonal projections of v_m onto each previous vector:

$$w_m = v_m - \sum_{j=1}^{m-1} \langle u_j, v_m \rangle u_j.$$

Because $v_m \notin \text{span}\{v_1, \dots, v_{m-1}\}$, $w_m \neq 0$. Normalize to obtain $u_m = \frac{w_m}{\|w_m\|}$.

Inductively, $\{u_1, u_2, \dots, u_k\}$ forms the required orthonormal set. ■

4.6.4 Geometric Interpretation of Length, Angle, and Projection

Standard inner products provide a precise bridge between vector algebra in $\mathbb{K}^n$ and classical geometry:

1. **Length (Norm):** $\|x\| = \sqrt{\langle x, x \rangle}$ represents the Euclidean distance from the origin to the point x .
2. **Distance Between Vectors:** The distance $d(x, y)$ between two vectors $x, y \in \mathbb{K}^n$ is defined by: $d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}$.
3. **Angle in $\mathbb{R}^n$:** For two non-zero vectors $x, y \in \mathbb{R}^n$, the Cauchy–Schwarz Inequality guarantees that $-1 \leq \frac{\langle x, y \rangle}{\|x\|\|y\|} \leq 1$. The unique **angle** $\theta \in [0, \pi]$ between x and y is defined by:

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\|\|y\|} \implies \theta = \arccos \left(\frac{\langle x, y \rangle}{\|x\|\|y\|} \right).$$

4. Orthogonal Projection: The orthogonal projection of a vector x onto a non-zero vector u is given by: $\text{proj}_u(x) = \frac{\langle u, x \rangle}{\|u\|^2} u$. The vector $x - \text{proj}_u(x)$ is perpendicular to u , decomposing x into orthogonal components parallel and perpendicular to u .

Example 4.6.9. Show that

$$\mathcal{U} = \left\{ u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is an orthonormal basis for $\mathbb{R}^3$, and express $x = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ using Fourier coefficients.

Solution. We check orthonormality by computing inner products $\langle u_i, u_j \rangle$:

$$\langle u_1, u_1 \rangle = \frac{1}{2}(1 + 1 + 0) = 1, \quad \langle u_2, u_2 \rangle = \frac{1}{2}(1 + 1 + 0) = 1, \quad \langle u_3, u_3 \rangle = 1.$$

$$\langle u_1, u_2 \rangle = \frac{1}{2}(1 - 1 + 0) = 0, \quad \langle u_1, u_3 \rangle = 0, \quad \langle u_2, u_3 \rangle = 0.$$

Thus, $\mathcal{U}$ is an orthonormal set of 3 vectors in $\mathbb{R}^3$, making it an orthonormal basis.

Now compute the Fourier coefficients $c_i = \langle u_i, x \rangle$:

$$\begin{aligned} c_1 &= \langle u_1, x \rangle = \frac{1}{\sqrt{2}}(1(3) + 1(1) + 0(4)) = \frac{4}{\sqrt{2}} = 2\sqrt{2}, \\ c_2 &= \langle u_2, x \rangle = \frac{1}{\sqrt{2}}(1(3) - 1(1) + 0(4)) = \frac{2}{\sqrt{2}} = \sqrt{2}, \\ c_3 &= \langle u_3, x \rangle = 0(3) + 0(1) + 1(4) = 4. \end{aligned}$$

Applying the Fourier expansion theorem: $x = 2\sqrt{2}u_1 + \sqrt{2}u_2 + 4u_3$. Verifying Parseval's identity: $\|x\|^2 = 3^2 + 1^2 + 4^2 = 26$, and $c_1^2 + c_2^2 + c_3^2 = (2\sqrt{2})^2 + (\sqrt{2})^2 + 4^2 = 8 + 2 + 16 = 26$. ■

Example 4.6.10. Apply the Gram–Schmidt process to transform the linearly independent set $\{v_1, v_2\} \subset \mathbb{R}^3$ into an orthonormal set $\{u_1, u_2\}$, where:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T.$$

Solution. Construct the first orthogonal vector $w_1 = v_1$:

$$w_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \implies \|w_1\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}.$$

Normalize w_1 to get u_1 :

$$u_1 = \frac{w_1}{\|w_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Construct the second orthogonal vector $w_2 = v_2 - \langle u_1, v_2 \rangle u_1$: First, compute the inner product:

$$\langle u_1, v_2 \rangle = \frac{1}{\sqrt{2}}(1(1) + 1(0) + 0(1)) = \frac{1}{\sqrt{2}}.$$

Then compute w_2 :

$$w_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}.$$

Compute the norm of w_2 :

$$\|w_2\| = \sqrt{(1/2)^2 + (-1/2)^2 + 1^2} = \sqrt{1/4 + 1/4 + 1} = \sqrt{3/2} = \frac{\sqrt{6}}{2}.$$

Normalize w_2 to get u_2 :

$$u_2 = \frac{w_2}{\|w_2\|} = \frac{2}{\sqrt{6}} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}.$$

The resulting orthonormal set is $\{u_1, u_2\}$. ■

Example 4.6.11. Let $x = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$ in $\mathbb{R}^3$. Calculate the distance $d(x, y)$, the angle θ between x and y , and the orthogonal projection $\text{proj}_y(x)$.

Solution. First, compute the individual norms and inner product: $\|x\| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$, $\|y\| = \sqrt{0^2 + 3^2 + 4^2} = \sqrt{25} = 5$, $\langle x, y \rangle = 1(0) + 2(3) + 2(4) = 0 + 6 + 8 = 14$.

To compute the distance $d(x, y) = \|x - y\|$:

$$x - y = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} \implies d(x, y) = \sqrt{1^2 + (-1)^2 + (-2)^2} = \sqrt{1 + 1 + 4} = \sqrt{6}.$$

To compute the angle θ :

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} = \frac{14}{3 \times 5} = \frac{14}{15} \implies \theta = \arccos\left(\frac{14}{15}\right) \approx 0.367 \text{ radians } (21.04^\circ).$$

To compute the orthogonal projection $\text{proj}_y(x)$:

$$\text{proj}_y(x) = \frac{\langle y, x \rangle}{\|y\|^2} y = \frac{14}{25} \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 42/25 \\ 56/25 \end{bmatrix}.$$
■

Exercise 4.6.12. Solve the following exercises.

1. Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \in M_2(\mathbb{R})$ and $p(x) = x^3 - 3x^2 + 2x - 1$. Evaluate the matrix polynomial $p(A)$.
2. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \in M_3(\mathbb{R})$. Show that A is nilpotent, state its index of nilpotency k , and evaluate $(I_3 - A)^{-1}$ and e^A as finite matrix polynomials.

3. Prove that if $A \in M_n(\mathbb{K})$ is an idempotent matrix ($A^2 = A$), then every eigenvalue of A is either 0 or 1.
4. Find the characteristic polynomial $p_A(\lambda)$ and the minimal polynomial $m_A(x)$ for the matrix:

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

5. Compute the characteristic polynomial $p_A(x)$ and minimal polynomial $m_A(x)$ for:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

6. Let $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix}$.

- (a) Find the characteristic polynomial of A .
 (b) List the possible minimal polynomials.
 (c) Determine the minimal polynomial by testing the candidates directly.

7. Find the eigenvalues, algebraic multiplicities, geometric multiplicities, and a basis for each eigenspace of:

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

8. Let $A, B \in M_n(\mathbb{K})$ be similar matrices ($B = P^{-1}AP$). Prove that A and B have the same characteristic polynomial, trace, determinant, and eigenvalues.
9. Prove that if $v_1, v_2, \dots, v_k \in \mathbb{K}^n$ are eigenvectors of a matrix A corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$, then $\{v_1, v_2, \dots, v_k\}$ is linearly independent.
10. Let $A \in M_n(\mathbb{K})$ be an invertible matrix. If λ is an eigenvalue of A with corresponding eigenvector x , prove that $\lambda \neq 0$ and that λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector x .
11. Verify the Cayley–Hamilton Theorem for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Use the theorem to compute A^{-1} and A^4 .
12. Verify the Cayley–Hamilton Theorem for the matrix:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix},$$

and use it to find A^{-1} .

13. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Using the Cayley–Hamilton Theorem, evaluate the high-degree matrix polynomial: $f(A) = A^5 - 2A^4 - 3A^3 + 2A^2 - 3A + 5I_2$.

14. Given $x = \begin{bmatrix} 2+i \\ 3 \\ -2i \end{bmatrix}$ and $y = \begin{bmatrix} 1-i \\ i \\ 2 \end{bmatrix}$ in $\mathbb{C}^3$:

- (a) Compute the complex inner product $\langle x, y \rangle = x^* y$.
- (b) Compute the norms $\|x\|$ and $\|y\|$.
- (c) Verify the Cauchy–Schwarz Inequality $|\langle x, y \rangle| \leq \|x\| \|y\|$.

15. Prove the Parallelogram Law for vectors in $\mathbb{K}^n$:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

16. Prove the Pythagorean Theorem in $\mathbb{K}^n$: if $x, y \in \mathbb{K}^n$ are orthogonal ($x \perp y$), then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.

17. Show that $S = \left\{ u_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, u_2 = \frac{1}{3} \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}, u_3 = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \right\}$ is an orthonormal basis for $\mathbb{R}^3$.

Find the Fourier coefficients of $x = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}$ relative to S .

18. Apply the Gram–Schmidt process to transform the linearly independent set $\{v_1, v_2, v_3\} \subset \mathbb{R}^3$ into an orthonormal set $\{u_1, u_2, u_3\}$, where:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

19. For the vectors $x = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ in $\mathbb{R}^3$:

- (a) Compute the distance $d(x, y) = \|x - y\|$.
- (b) Find the angle θ between x and y .
- (c) Compute the orthogonal projection $\text{proj}_y(x)$.

20. Prove Parseval's Identity: if $\{u_1, u_2, \dots, u_n\}$ is an orthonormal basis for $\mathbb{K}^n$, then for any vector $x \in \mathbb{K}^n$:

$$\|x\|^2 = \sum_{i=1}^n |\langle u_i, x \rangle|^2.$$

CONCLUDING NOTE TO STUDENTS

Congratulations!

“Mathematics is not about numbers, equations, or computations: it is about understanding.”

— William Paul Thurston

DEAR STUDENTS,

Congratulations on successfully completing the course on **Theory of Matrices (MMT126J)**! Reaching this milestone reflects your dedication, perseverance, and growing mathematical maturity.

Throughout this book, you have mastered the essential tools of linear algebra—transitioning from matrix algebra, structural symmetries, and elementary row operations (Units I–II) to linear independence, basis, rank, trace, spectral theory, the Cayley–Hamilton theorem, and inner product spaces (Units III–IV).

Matrix theory is the universal language of modern science, technology, engineering, and advanced mathematics. The conceptual clarity, abstract reasoning, and analytical problem-solving skills you have cultivated here will serve as an enduring foundation for your future studies and competitive examinations such as **CSIR NET-JRF, GATE, SET, IIT-JAM, and Ph.D. entrance tests**.

Keep exploring, stay curious, and carry this logical confidence forward into every domain of your academic journey. I wish you every success and fulfillment in all your future endeavors!

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